Computational Coupling

Four Choices Decide Whether a Connectome Beats Its Rewiring

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Abstract

Empirical connectomes are reported to outperform degree-preserving rewirings as reservoir computers — to have, in the term we use throughout, positive *computational coupling*: a score gap attributable to wiring and nothing else — and that result is increasingly invoked as a reason to build neuromorphic hardware on connectome-derived wiring. We find that the comparison it rests on is not one experiment but a composition of four choices — the null ensemble, the unit model, the operating point and the readout protocol — each normally reported in a subordinate clause, each able to change the sign of the answer on identical data, and so each a part of the coupling it reports.

The first is the one we got wrong. "Degree-preserving randomisation" names at least two ensembles, depending on whether an edge's weight travels with it or is reassigned afterwards; both preserve the degree sequence and the weight distribution exactly, and the phrase does not distinguish them. Over 2700 evaluations — 5 connectomes × 10 paired instances × 9 gains × two unit models — the two give the connectome's advantage opposite signs in 6 of 10 cells and opposite *significant* verdicts in 3 (median disagreement 0.73 capacity units, maximum 6.98). Against the ensemble the published toolbox builds the connectome leads in 1 of 5 networks; against the other, on the same graphs at the same operating point, in 5 of 5. The cause is that rank-reassignment *installs* a weight–degree association the data lacks (+0.66 median shift against +0.06), which predicts that the two must coincide on an unweighted network — and on the one binary network in the set they agree to 0.008 capacity units.

We do not read this as a refutation of the published finding — our harness differs in other respects too — but as evidence that a reproduction which builds its null from a published description cannot tell which hypothesis it tested. We report the remaining three choices on the same terms, together with a motif-preserving null ensemble, a gap-free perturbation bound whose prediction our own test partly falsifies, and a full audit of the withdrawn claims of an earlier version of this work.

Significance

Brains are wired in patterns that look far from random, and a widely repeated claim holds that those patterns are what make brains good at computing. How tightly a network's wiring is tied to what it can compute is what we call its computational coupling, and it cannot be read off a single network: it only shows up as a difference. The usual way to test for it is to take a real wiring diagram, shuffle its connections at random, and check whether the real one performs better at some task. If it does, the wiring pattern is credited with the difference.

We tried to check one such result and found that the test has a loose part. "Shuffle the connections at random" sounds like one procedure, but it is at least two, because a connection has both a location and a strength, and the phrase does not say whether a connection's strength should travel with it when it moves. Both readings are reasonable. We wrote one of them, believing we had written the other.

On the same wiring diagrams, measured the same way, the two shuffling procedures disagree about whether real brain wiring wins — not by a little, but in opposite directions, each with the statistical confidence normally taken as decisive. We can say why: one of the two procedures quietly creates a pattern that real brains do not have, which handicaps the comparison. That explanation makes a prediction we could have failed — on a wiring diagram where all connections have equal strength, there is nothing to move, so the two procedures must agree. One of our five networks is of that kind, and there they agree almost exactly.

Three further choices in the same test have the same character: what kind of model neuron is used, how strongly the network is driven, and which neurons are allowed to receive input and be read out. Each can reverse the answer on identical data, and none is usually reported as a choice at all.

None of this shows that brain wiring is unremarkable. It shows that computational coupling is not something a wiring diagram simply has — it is something a wiring diagram has relative to a way of asking, and a popular way of asking is less settled than it appears, and that the code which generates the comparison matters as much as the sentence describing it. We include our own mistake in full, because it produced a clean and convincing result that was answering the wrong question, and nothing in the output said so.

1 Introduction

Connectomes are now available at a scale that invites a structural explanation of neural computation: the complete somatic wiring of *C. elegans* [1], the adult *Drosophila* brain [2, 3], proofread mammalian cortical volumes [4], and whole-brain human networks [5]. Alongside them has grown a family of claims of the following shape: biological wiring contains statistically overrepresented subgraphs – feedforward loops, reciprocal inhibitory pairs – these motifs implement identifiable computations, and networks built to contain them inherit an advantage in accuracy, robustness or energy over conventional artificial networks.

Each step of that chain is plausible, and the first step is well established in the sense that motif overrepresentation against degree-preserving nulls is a reproducible empirical finding [6–8] — our own recomputation finds it too, though less uniformly across species than the phrase "conserved motifs" suggests (§3.1). The later steps are where the evidence is thinner than the claims, for a reason that is methodological rather than empirical.

A contradiction, and what it turned out to be about. The sharpest quantitative support for the later steps comes from reservoir computing, and it is the part of this literature that reaches furthest: it is routinely offered as a reason to build neuromorphic systems on connectome-derived wiring. Suárez et al. [9] report that empirical connectomes have significantly greater linear memory capacity than degree-preserving rewirings. Our own ladder, run in a spiking network at a higher gain, found no such advantage.

We set out to adjudicate that disagreement and did not manage to. What we found instead is that the comparison "does a connectome beat its rewiring?" is not one experiment. It is a composition of four choices, each of which is normally reported in a subordinate clause, and each of which can change the sign of the answer on identical data.

Computational coupling. We call the quantity this literature is trying to measure the *computational coupling* of a network: the extent to which what a network computes is attributable to how it is wired. Coupling is never observed directly. What is observed is a score on a task, and a score belongs to the network together with everything the measurement supplies around it — the units the network is built from, the regime it is driven in, the channels it is read through, and the alternative networks it is scored against. The coupling is the part of that score which moves when the wiring moves and nothing else does. It is therefore defined only relative to a stated procedure for moving the wiring, and it is the sign of that difference, not the score itself, that the structural claims in this literature rest on.

Definition 1 (Computational coupling)**.** *Let $C_{U,\rho,R}(W)$ denote the task capacity of a network with weight matrix W, evaluated with unit model U, at operating point ρ, under readout protocol R; and let $\mathcal{N}$ be an ensemble of surrogate networks that alters the wiring while holding stated properties of W fixed. The* computational coupling *of W under the measurement protocol $P = (\mathcal{N}, U, \rho, R)$ is the standardised contrast*

$$\kappa(W; P) \;=\; \frac{C_{U,\rho,R}(W) \;-\; \mathbb{E}_{W'\sim\mathcal{N}}\, C_{U,\rho,R}(W')}{\mathrm{sd}_{W'\sim\mathcal{N}}\; C_{U,\rho,R}(W')}\,. \qquad (1)$$

Equation 1 is not a new statistic: it is the Z-score the field already reports, under connectome-versus-null comparisons, motif enrichment, small-worldness and rich-club analysis alike. Naming it is worth the trouble for one reason — the name carries its arguments with it. "The connectome beats its rewiring" is the claim $\kappa > 0$, and κ is a function of four things besides W. Each of $\mathcal{N}$, U, ρ and R is a choice; each is normally reported in a subordinate clause, if at all; and each, we find, can change the sign of κ on identical data. A coupling reported without its protocol is not a measurement of the connectome, because the same connectome supports both signs. We measured all four choices (Table 1).

1. **Which rewiring counts as the null** ($\mathcal{N}$)**.** "Degree-preserving randomisation" names at least two different ensembles, depending on whether a weight travels with its edge or is re-assigned afterwards. Both match the phrase; they are not the same distribution over graphs; and on the same connectomes, at the same operating point, in the same unit model, they do not agree about the sign of the result (§3.6).

2. **What a unit is** (U)**.** A tanh unit is never silent; a subthreshold integrate-and-fire unit contributes nothing at all. Running identical graphs, identical rewirings, identical input and an identical readout through both — on the five connectomes distributed with the original toolbox — changes the outcome, and it matters because the hardware the result is invoked for is spiking (§3.7).

3. **Where on the gain axis it is measured** (ρ)**.** The published comparison is located *at criticality*, at a spectral radius of 1, the regime a reservoir is conventionally evaluated in [10–12]. The connectome-versus-null contrast changes sign along that axis — in 3 of 5 networks, significant on both sides — so a comparison reported at one gain is a statement about that gain (§3.8).

4. **How the reservoir is read out** (R)**.** Under a protocol in which every neuron is both driven and decoded, a reservoir with its recurrent matrix *set to zero* scores 59.6% against 59.6% for the connectome. In that regime the comparison is not measuring the wiring at all (§3.3).

This is the paper's central claim, and it is a claim about how the field measures rather than about what brains are. Computational coupling is a property of a connectome *and* the apparatus that measures it, and the four arguments of κ are that apparatus. Each of the four is a fact about the

Table 1: **Four choices that a connectome-versus-null comparison has to make, and what each one is worth.** Each row is a decision normally reported in a subordinate clause of a methods section. For each we ran both settings on otherwise identical data and report the largest effect the choice alone produces. None of the four is a robustness check in our design: each can change the sign of the reported result, and the first and fourth can change it between two significant verdicts in opposite directions.

Choice	Setting A	Setting B	What the choice alone does
Null ensemble (§3.6)	BCT `randmio`: weights travel with their edges	Rank-reassigned: same weight multiset, re-placed by degree rank	Opposite signs in 6 of 10 cells; opposite significant verdicts in 3; median disagreement 0.73 capacity units
Unit model (§3.7)	Graded tanh, never silent	Leaky integrate-and-fire, silent below threshold	Capacity compressed 4.9× for every topology alike; the contrast survives in 0 of 5 networks
Operating point (§3.8)	At criticality, $\rho = 1$	Above it, $\rho > 1$	Contrast changes sign along the axis in 3 of 5 networks, significant on both sides
Readout protocol (§3.3)	All-to-all: every neuron driven and decoded	Sensory/motor-disjoint	Under A, zeroing the recurrent matrix entirely costs nothing (59.6% against 59.6%); under B it falls to chance (4.9%)

measurement that is indistinguishable, in a single reported number, from a fact about the topology. The fourth is the limiting case: under an all-to-all readout the coupling is driven to zero by construction, since a reservoir whose recurrent matrix has been deleted outright scores as well as the connectome, and a comparison run there reports a number that no change to W could have moved. Our own results carry exactly the same dependence, which is why we report the axis and not the point for them too — and why, where our design could not resolve an effect, we say so rather than reporting a non-detection as a refutation.

The attribution problem. Establishing that a structural property *causes* a computational advantage requires a control that differs in that property and nothing else. The standard comparison – a connectome against a degree-preserving rewiring, or against an Erdős–Rényi graph – does not have this form. Rewiring a connectome simultaneously changes its motif census, its clustering, its spectral structure and every higher-order statistic that covaries with them, so a performance difference cannot be attributed to motifs in particular. The ensemble that would settle the question — one that randomises a connectome while *holding its motif census fixed* — is standard in statistical network science and absent from this debate. Motif-constrained ensembles have been sampled by annealing [6], by exponential random graph models [13, 14], by the *dk*-series [15, 16], and, with a sampling guarantee, by Wang–Landau multicanonical methods [17]. None has been used to ask what a connectome's motifs *do*. We build a sign-attributed version at connectome scale and place it in a nested ladder of controls in which each rung adds exactly one structural property, so that the contribution of each can be read off as a difference between adjacent rungs.

Three further confounds recur in this literature and are worth stating plainly, since all three appear in our own earlier work on this project. First, a null model described as preserving degrees *and* Dale's principle preserves nothing beyond degrees: when synapses inherit their presynaptic neuron's valence, Dale's principle is automatically invariant under double-edge swaps, so such a null cannot test whether the placement of inhibition carries information. Second, and more consequential, a reservoir in which every neuron is both driven and decoded does not require its recurrent connectivity at all – we show that zeroing the recurrent matrix entirely leaves SHD accuracy unchanged under that protocol. A structural comparison run in that regime measures nothing, whatever it reports. Third, matching spectral radius across networks equalises recurrent gain but not the activity level each network settles into, and since firing rate both predicts accuracy and determines the synaptic-operation count outright, a comparison at matched gain confounds topology with operating point – in the direction that flatters the unstructured control. We therefore report the ladder both at matched gain and with every network recalibrated to a common firing rate.

A common thread runs through all three: each is a failure that produces no error and no warning. The naive protocol returns plausible accuracies, the Dale-compliant null returns plausible Z-scores, and the matched-gain comparison returns a plausible ordering. In the first two the object of study turns out to have made no difference to the number reported; in the third it made one, but not for the reason the comparison was set up to attribute it to. Nothing in any of their outputs says so. That is why we regard the controls that expose them as part of the method rather than as robustness checks.

The degeneracy problem. A second strand asks the converse question: how much can a circuit be rewired before function degrades? Biological circuits are famously degenerate [18–20], and the natural formal tool is the Davis–Kahan theorem [21], bounding invariant-subspace rotation by the perturbation size over the spectral gap. Applied to measured connectomes the bound is vacuous, because their spectral gaps are tiny (0 at the smallest): it permits a perturbation of norm at most 2.2×10^{-2} while the networks tolerate a great deal more. Reporting the resulting discrepancy as a finding about biology inverts the situation – it is a statement about the instability of eigenvectors under near-degenerate spectra. We replace it with a bound on the object that task performance actually depends on, the finite-horizon input–output map, which requires no gap assumption; and we extract from it a quantity, the operational alignment γ of a perturbation, that turns "biological rewiring is non-adversarial" into a measurable prediction. Crucially, the prediction is falsifiable, and it is partly falsified: we construct the maximally damaging perturbation of matched cardinality and find that γ ranks the perturbation directions by tolerance as predicted, but does not reproduce the $1/\gamma$ scaling the bound implies (§3.11). We report the failure alongside the part that holds. What survives is enough for the point at issue: a degeneracy result demonstrated only against random pruning shows that random directions are benign, which is a much weaker statement than it appears.

Contributions.

1. An exact sign-attributed motif census with $O(d)$ incremental updates, resting on the observation that under Dale's principle a feedforward loop's coherence is decided solely by the valence of its intermediate neuron (Eq. 2).
2. A **motif-preserving null ensemble**: degree-, valence- and census-matched surrogates sampled by an annealed swap chain. On *C. elegans* it matches all six targeted motif counts to within 0.3% while retaining only 13.9% edge overlap with the empirical graph, against 11.6% for unconstrained rewiring, and it converges on every network we test — including diffusion tractography, where almost every edge is reciprocal and an ordinary swap chain cannot hold the census at all.
3. A **sampler-robustness analysis** of that ensemble: the same surrogates drawn over a factor-of-eight range of chain lengths and from both possible starting points (the empirical graph under a census constraint, and a fully rewired graph annealed towards the census), which rules out the two ways an annealed sampler could manufacture the null result reported below.
4. A **structural decomposition** of reservoir performance on a real spike-based benchmark, isolating the contribution of density, degree sequence, valence placement, motif census and residual wiring, with matched controls, effect sizes and multiplicity correction.
5. A **null-ensemble sensitivity analysis**: two rewiring constructions that both satisfy the phrase "degree-preserving randomisation" — one carrying weights with their edges, one re-assigning them afterwards — run over the full design, giving the connectome's advantage opposite significant verdicts in 3 of 10 cells. We built the second believing it was the first, and report the episode as the worked example it is rather than removing it.
6. A **unit-model comparison** on the connectomes distributed with `conn2res`: identical graphs, rewirings, inputs, delays and readout run through graded and spiking units, with capacity compressed 4.9× between them, and with the silencing hypothesis we proposed for the difference supported against the published null after we had reported it as refuted against the other one.
7. A **sweep of the gain axis** showing that the connectome-versus-null contrast changes sign along it, and that against an unstructured graph it does not appear at any gain we tested.
8. A **gap-free perturbation bound** on the transfer map, the operational alignment γ, and an adversarial-pruning test of the resulting prediction, which it passes ordinally and fails quantitatively.
9. A **reproducible open-PDK synthesis** of the accompanying RTL, reporting only quantities the flow can establish, and withdrawing previously reported commercial process figures that were never measured.

We regard the fifth item, and the corrections it entails, as part of the contribution rather than an appendix to it. The audit record accompanying this work documents each withdrawn claim and the evidence against it.

2 Methods

2.1 Connectomes, and what each one can support

We analyse four public connectome releases spanning three phyla, from which we take five networks — the *Drosophila* release supplies two functionally distinct circuits (Table 2). Two of the releases carry synaptic valence and two do not, and this distinction constrains which questions each dataset can answer; we state it here rather than in a caption because it is the reason several rows of our tables are deliberately empty.

Caenorhabditis elegans. The adult hermaphrodite somatic connectome [1, 22], with neurotransmitter assignments giving an excitatory/inhibitory (E/I) label per neuron.

Drosophila melanogaster. FlyWire v630 [2, 3], from which we take the central complex and the antennal lobe as two functionally distinct circuits. Valence follows the released neurotransmitter predictions.

Mus musculus. The MICrONS Pinky100 V1 layer 2/3 proofread subgraph [4]. This release annotates 34 inhibitory somata, but every one of them is disconnected in the accompanying synapse table: all 1,736 synapses are excitatory→excitatory. We therefore treat this network as **topology-only** and do not compute sign-attributed motifs on it. Reporting coherent feedforward loops here would be reporting plain feedforward loops under another name, since with no inhibitory neuron in the graph the two counts are identically equal.

Homo sapiens. The Human Connectome Project group-consensus structural network [5], 1,015 parcellated regions from diffusion tractography. Streamline density carries no synaptic valence, so this network is likewise treated as topology-only. We emphasise that it is a regional network, not a neuronal connectome, and we make no microscale claim from it.

We do not analyse human microscale connectivity. The H01 release [23] provides cell and synapse tables but the connectivity export needed to build a wiring diagram is not part of the distribution we worked from, and a census of a synapse table is not a connectome analysis.

Reservoir capacity grows with neuron count, so cross-species comparison requires size matching. From each connectome we extract an induced subgraph of at most $N = 400$ neurons by breadth-first expansion from a seed neuron of above-median degree, following both afferent and efferent edges; degree-zero nodes are then removed. Uniform node sampling is not used, because it destroys degree structure and motif density by construction.

2.2 Exact sign-attributed motif census

For a directed graph with neuron valences $\sigma_i \in \{+1, -1\}$ and Dale's principle – every synapse carries the valence of its presynaptic neuron – the sign product around a feed-forward loop $(u \to v,\ v \to w,\ u \to w)$ is

$$\sigma_u \cdot \sigma_v \cdot \sigma_u \ = \ \sigma_u^2\, \sigma_v \ = \ \sigma_v. \tag{2}$$

Remark 1. *A feedforward loop is coherent if and only if its* intermediate *neuron is excitatory. Coherence is a property of one node, not of a product over three edges.*

Equation (2) makes the sign-attributed census exactly as cheap as the topological one and removes any need to store per-edge signs. We count

$$\mathrm{FFL} = \sum_{(u,v)\in E} \big|N^+(u) \cap N^+(v)\big|, \tag{3}$$

$$\mathrm{cFFL} = \sum_{\substack{(u,v)\in E \\ \sigma_v = +1}} \big|N^+(u) \cap N^+(v)\big|, \tag{4}$$

with $\mathrm{iFFL} = \mathrm{FFL} - \mathrm{cFFL}$; reciprocal pairs, mutually inhibitory pairs and directed 3-cycles follow analogously, and bi-fans are counted exactly by accumulating source-pair co-occurrence through shared targets rather than by sampling node pairs. All counts are exact; none are estimated. Single-edge insertions and deletions admit exact $O(d)$ updates of every 3-node count, which is what makes the annealed null model of §2.3 tractable.

2.3 Null ensembles and the structural control ladder

Dale's principle constrains nothing here. The conventional null is the Maslov–Sneppen double-edge swap [24], which fixes the joint in/out degree sequence. It is common to describe a connectome null as preserving degrees *and* Dale's principle. Under the convention above this is not an additional constraint: the swap $(u \to v,\ x \to y) \Rightarrow (u \to y,\ x \to v)$ leaves u and x presynaptic, so every neuron's outgoing synapses retain a single shared valence automatically. A "Dale-compliant" rewiring is therefore the standard ensemble, and it cannot test whether the *placement* of inhibition in the graph is non-random. That requires permuting valence labels across neurons, which we include as a separate rung.

Two weighted rewirings, and which one we report. For the weighted `conn2res` connectomes the swap above underdetermines the ensemble, because it says what happens to edges and not what happens to weights. We implement both readings and report both (§3.6).

Weight-carrying rewiring moves each weight with the edge that carries it, and rejects swaps that disconnect the graph. We obtain it by calling `bct.reference.randmio_und_connected` or `randmio_dir_connected` [25] directly, dispatching on symmetry and using the toolbox default of ten swap attempts per edge, rather than reimplementing it — the point of §3.6 is that a reimplementation from a description is not guaranteed to be the same ensemble. This is what `conn2res` calls, so it is the ensemble every comparison in this paper reports unless stated otherwise.

Rank-reassigned rewiring randomises the binary topology by the same swaps and then assigns the original multiset of weight values to the new edges in decreasing order of endpoint-degree product $d_u^{\mathrm{out}} d_v^{\mathrm{in}}$. It preserves the degree sequence and the weight distribution exactly and destroys the association between a weight and its edge. Both constructions satisfy the description "randomised while preserving density and degree sequence"; they are different distributions over graphs; and §3.6 reports what that difference is worth.

Both are applied *after* the toolbox's preprocessing (min–max scaling followed by division by the spectral radius) and before the gain is set, so that each arm of a comparison is a randomisation of the same object.

The motif-preserving ensemble. The rung the connectome literature does not use is one that preserves the motif census itself. Constructions for it exist outside that

literature and we do not claim the idea: Milo et al. proposed a simulated-annealing null preserving lower-order subgraph counts [6, 7]; exponential random graph models condition on triad counts by design [13, 14]; the *dk*-series is a nested hierarchy of nulls constraining progressively higher-order structure, with $2K$ and $3K$ fixing degree correlations and connected triads [15, 16]; and Fischer et al. sample motif-constrained ensembles at fixed degree sequence by Wang–Landau multicanonical sampling, which unlike annealing comes with a sampling guarantee [17]. What we contribute is an implementation for the *sign-attributed* census at connectome scale — exact coherent and incoherent feedforward-loop counts under Dale's principle, with $O(d)$ incremental updates and a move set that survives near-total reciprocity — and the use of that ensemble as a control rung rather than as an enrichment null. Where a formally exact sampler is available it should be preferred, and §3.9 reports what our annealed chain can and cannot demonstrate in its place. We sample it with a Metropolis chain over degree-preserving swaps whose energy is the squared deviation of the running census from a target census c^*, measured in units of the unconstrained-ensemble standard deviation s_m:

$$\mathcal{E}(G) \;=\; \sum_{m\in\mathcal{M}} \left(\frac{c_m(G) - c^*_m}{s_m} \right)^2, \qquad (5)$$

over $\mathcal{M}$ = {reciprocal pairs, mutually inhibitory pairs, FFL, cFFL, iFFL, 3-cycles}, over $120\times|E|$ proposals with a geometric temperature schedule. Exact incremental updates (§2.2) make each proposal $O(d)$. Setting c^* to the empirical census yields graphs that are maximally randomised subject to reproducing the connectome's motif spectrum.

Two things about the construction are worth stating precisely, because the obvious implementation of it does not work. First, where the chain starts determines which property is guaranteed and which must be demonstrated. Starting from a fully rewired graph and annealing towards c^* guarantees randomisation and leaves convergence to be shown; starting from the connectome itself and running a low-temperature constrained walk guarantees convergence — the chain begins on the constraint surface — and leaves randomisation to be shown. These are equivalent where both succeed, but the random start fails outright on networks whose census is hard to rebuild by chance. We therefore use the empirical start and demonstrate randomisation directly, by edge overlap against the empirical graph and against an unconstrained rewiring (§3.2).

Second, an ordinary double-edge swap destroys reciprocal pairs far more readily than it creates them. In a near-symmetric network — diffusion tractography above all, where almost every edge is reciprocal — the reciprocal-pair count therefore drifts down by tens of percent however stiff the penalty, and the constrained ensemble is effectively unreachable by that move set. We add a move that swaps two reciprocal *pairs* as units, $(u \leftrightarrow v,\, x \leftrightarrow y) \mapsto (u \leftrightarrow y,\, x \leftrightarrow v)$, which preserves every node's in- and out-degree and the reciprocity of all four edges, and mix it in proportion to the fraction of edges that are reciprocal. Bi-fan is deliberately excluded from $\mathcal{M}$ and reported post hoc as a held-out statistic, which tests whether the chain reproduces structure it was not asked to reproduce.

The ladder. Each rung adds exactly one structural property to the one below it, so the contribution of that property is the difference between adjacent rungs:

L0 Erdős–Rényi: node count, edge count and E:I ratio only.

L1 + joint in/out degree sequence (valence labels permuted across nodes).

L2 + valence placement (labels left on their own neurons).

L3 + 3-node motif census (the annealed ensemble above).

L4 the empirical connectome.

Every rung has identical node and edge counts by construction. We draw 15 independent surrogates per rung. Rung L4 is different in kind: the connectome is one graph, so its 15 replicates differ in the input coding and the sensory/motor split only. Its replicates therefore carry one variance component where L0–L3 carry two, and we state the consequence rather than leave it to be inferred. Contrasts use Welch's test, which does not assume equal variances, but a contrast involving L4 supports inference about *this* connectome rather than about connectomes as a class; the surrogate rungs support the stronger reading. The asymmetry works against us where our conclusion is a null one: in the L3–L4 contrast a real difference specific to the empirical graph would be easier, not harder, to detect at reduced variance, so a small measured effect there is not an artefact of the variance structure.

2.4 Task and evaluation protocol

Benchmark. We use the Spiking Heidelberg Digits (SHD) dataset [26]: 20 spoken-digit classes in English and German, converted to spike trains by a cochlear model, from 12 speakers, with a canonical test split that holds out two speakers entirely. Spikes are binned into 100 time steps over 1 s and the 700 cochlear channels are pooled into 140 bands. We use a real spike-based benchmark rather than a synthetic pattern task because synthetic tasks of the kind used in earlier work saturate near 100% accuracy for every architecture and therefore cannot discriminate between them.

Reservoir protocol. The connectome or surrogate supplies the recurrent weight matrix and is never trained; only a ridge readout is fitted [9, 27]. Neurons are discrete-time leaky integrate-and-fire units,

$$v_i(t{+}1) = \lambda\, v_i(t)\,\big(1{-}s_i(t)\big) + I_i(t), \qquad s_i(t) = \mathbb{I}[v_i(t) \geq \vartheta]\,, \qquad (6)$$

with $I(t) = x(t)W_{\text{in}} + s(t)W_{\text{rec}}$, leak $\lambda = 0.90$ and threshold $\vartheta = 1$.

Controls that the comparison depends on. Four quantities are held fixed across every network in a comparison: node count, edge count, spectral radius (each W_{rec} is rescaled to $\rho = 1.30$), and the input projection. Without spectral-radius matching the comparison measures effective gain rather than topology.

A fifth control is less obvious and turned out to be decisive. If every neuron receives the stimulus and every neuron feeds the readout, then the task is solved by the input projection and the per-neuron leak alone: under that protocol a reservoir with $W_{\text{rec}} = 0$ scores 59.6% on SHD, statistically indistinguishable from any connectome, and no structural comparison can detect anything. We therefore drive a sensory subset (15% of neurons) and decode from a *disjoint* motor subset (35%), so task-relevant signal must traverse the recurrent graph to be decodable. The zero-matrix control then falls to 4.9% against a chance level of 5.0%, which is the diagnostic that the benchmark is sensitive to recurrent structure at all. We recommend that any structural comparison of this kind report the corresponding control.

Matching the operating point, not only the gain. A sixth control is needed because spectral radius fixes the recurrent gain but not the activity level the network settles into. At the common $\rho = 1.30$ the rungs do not fire at the same rate: the Erdős–Rényi surrogates run at roughly twice the mean rate of the connectomes they are matched to (§3.5), and firing rate is both correlated with accuracy across our runs ($r = 0.38$) and the direct determinant of the synaptic-operation count used as an energy proxy. A contrast at matched ρ therefore confounds topology with operating point, and in the direction that flatters the unstructured control.

We consequently report the ladder twice. In the *matched-gain* version every network is set to $\rho = 1.30$, and the rung differences include whatever change in activity level the topology induces. In the *matched-rate* version each network is calibrated individually, by bisection on its own spectral radius, to a common mean firing rate of 0.07, so that the comparison is made at a common operating point and the residual difference is attributable to wiring rather than to activity. The two versions answer different questions — at equal gain, what operating point does each topology select; and at equal operating point, does topology still change what the network computes — and we regard reporting only the first as the more common error.

Readout and statistics. Features are the time-averaged firing rate and the final membrane potential of the motor population. A ridge readout is fitted with the regulariser selected on a held-out validation split, never on test. Energy is reported as synaptic operations, $\mathcal{S} = \sum_t \sum_{i:s_i(t)=1} k_i^{\text{out}}$. Comparisons between rungs use Welch's t-test with Holm correction [28] across the full family of contrasts, and are reported with percentile-bootstrap confidence intervals and Hedges' g, because with 15 instances per rung a detectable difference can still be operationally negligible.

2.5 Perturbation theory without a spectral gap

Degeneracy is naturally posed as matrix perturbation, and the standard tool is the Davis–Kahan $\sin\Theta$ theorem [21, 29], which bounds the rotation of an invariant subspace by $\|E\|/\delta$ for a spectral gap δ. On real connectomes this is vacuous: the smallest measured gap is 0 (§3.11), so the bound admits perturbations of norm at most 2.2×10^{-2} — far smaller than the removal of a single synapse — while the networks in fact tolerate removal of a large fraction of their edges. That ratio is a statement about eigenvector instability under near-degenerate spectra, not about brains.

What must be stable is not an eigenvector but the finite-horizon input–output map. For horizon T that map is the sequence of Markov parameters $H_t = W_{\text{out}} A^t W_{\text{in}}$, $t = 0, \dots, T-1$.

Proposition 1 (Gap-free sensitivity of the transfer map)**.** *Let $a = \|A\|_2$. For a perturbation $A \mapsto A + E$, to first order in E,*

$$\|\Delta H_t\|_F \;\le\; t\, a^{\,t-1}\, \|E\|_F\, \|W_{\text{in}}\|_2\, \|W_{\text{out}}\|_2, \tag{7}$$

and hence, aggregating over the horizon,

$$\Big(\textstyle\sum_{t<T} \|\Delta H_t\|_F^2\Big)^{1/2} \;\le\; C(T,a)\, \|E\|_F, \tag{8}$$

$$C(T,a) = \|W_{\text{in}}\|_2 \|W_{\text{out}}\|_2 \Big(\textstyle\sum_{t<T} t^2 a^{2(t-1)}\Big)^{1/2}. \tag{9}$$

No spectral gap is assumed, and C is finite for every A.

Proof. Expand $(A+E)^t - A^t = \sum_{s<t} A^s E A^{t-1-s} + O(\|E\|^2)$ and bound each of the t terms submultiplicatively by $\|A\|_2^s\, \|E\|_F\, \|A\|_2^{t-1-s} = a^{\,t-1}\|E\|_F$; multiply by the input and output operators and collect. □

Remark 2. *The constant is stated in the operator norm and not in the spectral radius. Substituting $\rho(A)$ for $\|A\|_2$ would require $\|A\|_2 \le \rho(A)$, which holds only for normal matrices; connectome matrices are markedly non-normal, and we measure $\|A\|_2/\rho(A)$ up to 2.1 across our networks at matched ρ. With ρ in place of a the constant is too small by up to that ratio raised to the horizon, and the measured alignment γ — which is defined as the fraction of the worst case a direction realises — can exceed 1. This matters because transient amplification in a non-normal recurrent network is precisely the regime the bound is meant to cover.*

Proposition 1 is worst-case over the *direction* of E. Real remodelling is not adversarial, and the quantity that expresses this is the fraction of the worst case a given direction realises.

Definition 2 (Operational alignment)**.**

$$\gamma(E) \;=\; \frac{\big(\sum_{t<T}\|\Delta H_t\|_F^2\big)^{1/2}}{C(T,a)\,\|E\|_F} \;\in\; [0,1]. \qquad (10)$$

Proposition 2 (Tolerance scales inversely with alignment)**.** *If task performance degrades once the aggregated transfer-map perturbation exceeds a task-determined tolerance τ, then the largest admissible perturbation magnitude satisfies $\|E\|_F^{\max} \geq \tau/\big(\gamma(E)\,C(T,a)\big)$, i.e. it scales as $1/\gamma$.*

This is falsifiable in a way the usual degeneracy demonstration is not. Showing that a connectome survives *random* pruning establishes only that random directions are benign. We therefore construct the maximally damaging perturbation of the same cardinality by ranking synapses on their exact first-order influence on the horizon-aggregated map,

$$\frac{\partial}{\partial A_{ij}}\sum_{t<T}\big\|W_{\text{out}}A^tW_{\text{in}}\big\|_F^2 = 2\sum_{t<T}\sum_{s<t}\Big[\big(W_{\text{out}}A^{\,t-1-s}\big)^\top H_t\big(A^sW_{\text{in}}\big)^\top\Big]_{ij}, \qquad (11)$$

evaluated by accumulating forward states A^sW_{in} and adjoint states $W_{\text{out}}A^j$ iteratively, so no matrix power is ever formed. If a connectome tolerated adversarial and random pruning equally, Proposition 2 would be false. Every pruned network is renormalised to the original spectral radius before evaluation, so measured degradation reflects lost structure rather than lost gain, and γ is computed on exactly the perturbation applied.

2.6 Synthesis flow

The accompanying Verilog is synthesised with Yosys [30] and mapped with ABC [31] onto the Nangate45 Open Cell Library [32] at the typical corner. We report gate count, flip-flop count and mapped pre-layout cell area, and nothing else. We do not report die area, power, energy per operation or a clock frequency: this flow performs no placement, routing, clock-tree synthesis or parasitic extraction, switching power cannot be estimated without activity factors and extracted parasitics, and the toolchain ships no static timing analyser, so a critical path would have to be guessed rather than measured. Earlier revisions of this project reported such figures for a commercial 28 nm flow; no such flow was run, and those figures are withdrawn (see the repository's audit record).

3 Results

Table 3 lists every experiment reported below with its size, counted from its own output file. We give it up front because the argument of this paper is cumulative — no single comparison in it is decisive, and several of them contradict each other by design.

Table 2: **Size-matched subcircuits and the analyses each supports.** Networks are induced subgraphs grown by breadth-first expansion and stripped of degree-zero nodes. The final column records whether synaptic valence is available: in the MICrONS release used here every annotated inhibitory soma is disconnected, and diffusion tractography carries no valence at all, so neither network can support sign-attributed motif analysis.

Network	Species	N	$\lvert E\rvert$	Dens. (%)	$\bar{k}$	Inh. (%)	Valence
C. elegans	C. elegans	300	5,044	5.62	16.8	8.7	yes
Drosophila CX	D. melanogaster	400	7,856	4.92	19.6	45.5	yes
Drosophila AL	D. melanogaster	400	16,468	10.32	41.2	29.8	yes
Mouse V1 L2/3	M. musculus	334	1,734	1.56	5.2	0.0	**no**
Human HCP	H. sapiens	400	6,092	3.82	15.2	0.0	**no**

3.1 Motif enrichment, recomputed

We first re-establish the empirical starting point on the size-matched subcircuits of Table 2, using $K = 500$ degree-preserving surrogates per network (resolution floor $p \geq 0.0020$) and exact counts throughout.

Enrichment is real and often large, but it is not uniform, and the recomputation does not support the description of these motifs as conserved across phylogeny. Of the 29 interpretable (network, motif) tests, 23 are significantly enriched, 3 are significantly *depleted*, and 3 are indistinguishable from the degree-preserving null.

Figure 1 shows the pattern. In *C. elegans* every motif is strongly enriched: feedforward loops reach $Z = +92.1$ and reciprocal pairs $Z = +100.9$. The *Drosophila* central complex breaks the pattern in the coordinate the motif literature cares about most — its coherent feedforward loops are depleted, $Z = -10.9$, against an enrichment of $+84.4$ in the worm — while its reciprocal pairs remain enriched ($Z = +37.9$) and its directed 3-cycles are depleted ($Z = -6.2$). Mouse V1 L2/3 is the clearest case: no motif in that network is enriched by more than $Z = +2.7$, and directed 3-cycles ($Z = +0.6$) and bi-fans are indistinguishable from chance. Correspondingly, the normalised significance profiles of the two sign-attributed fly circuits agree with each other no better than either agrees with the worm (0.52 for fly–fly, against 0.55 and 0.91 for worm–fly), so what is conserved is not a profile shape but the presence of some excess somewhere.

Four qualifications belong with these numbers rather than in a footnote. First, sign-attributed motifs are reported only for *C. elegans* and *Drosophila*; for the mouse and human networks the coherent and total feedforward loop counts would be equal by construction (§2.1), so quoting a coherent-FFL enrichment there reports a topological result under a biological name. Second, the enrichment is measured against a degree-preserving ensemble, and because Dale's principle is automatically preserved by double-edge swaps, no part of this enrichment can be attributed to excitatory/inhibitory organisation. Third, the very large Z-scores in the human row ($Z = +169.3$ for feedforward loops) should not be read as a stronger biological result than the invertebrate ones: HCP tractography

Table 3: **Every experiment in this study.** Counts are read from each experiment's own output file at build time rather than transcribed, so the table cannot disagree with what was run; an experiment whose output is absent is reported as absent. This list is the same one the accompanying documentation prints, generated from a single source. The unit differs by row — a "run" is one network at one rung trained to convergence, an "evaluation" one network at one operating point probed for one metric — so the totals are not comparable across rows and are not summed.

Section	Experiment	Size
§3.1	Are the classic subgraphs really over-represented? Exact sign-attributed counts against 500 degree-preserving surrogates per network.	35 tests
§3.2	Validation of the annealed chain: does it hit the target census, does it actually randomise, and does it leave untargeted structure alone?	5 networks
§3.9	The same ensemble redrawn over a factor-of-eight range of chain lengths and from both possible starting points, ruling out two ways a sampler could fake the result.	200 draws
§3.3	What does the benchmark measure? Includes the reservoir with its recurrent matri× set to zero, under both readout protocols.	8 conditions
§3.4	L0 density to L4 real connectome on spoken digits, each rung adding exactly one structural property; accuracy, memory capacity, kernel and generalisation rank.	375 runs
§3.5	The same decomposition with every network recalibrated to a common firing rate, because matching spectral radius does not match activity.	250 runs
§3.4	Does taking a size-matched subgraph of a larger connectome bias the degree result? Run because the ladder depends on the extraction being neutral.	5 networks
§3.6	The headline experiment. Both readings of "degree-preserving randomisation" against both unit models across nine gains, on the original toolbox's own connectomes.	2,700 evaluations
§3.6	Structural descriptors of both ensembles against the real graph, including the prediction that they must coincide on an unweighted network.	105 graphs
§3.7	Input gain × threshold × leak, searching for the configuration that would refute our own spiking result rather than the one that confirms it.	1,728 evaluations
§3.8	The whole decomposition across the critical region, because a comparison at one gain is a statement about that gain.	1,400 evaluations
§3.8	Locating the published advantage on the gain axis at matched spectral radius.	2,500 evaluations
§3.10	The same decomposition measured on task-independent probes, so the ordering does not depend on one benchmark.	250 runs
§3.11	Random, hub-targeted and adversarial edge removal, testing whether the operational alignment predicts how much damage a direction does.	540 prunings
§3.12	Real logic synthesis and technology mapping of the accompanying RTL, reporting only what the open flow can establish.	4 designs

is an essentially undirected network, in which almost every edge is reciprocal, and a degree-preserving null of such a graph is a poor match for it in ways that have nothing to do with computation. Fourth, and most importantly for what follows: overrepresentation is a statement about a graph, not about a computation. The remainder of the paper asks what, if anything, it buys.

3.2 The motif-preserving ensemble

Table 4 reports the three properties the annealed ensemble has to have simultaneously. The chains reach the empirical census to within 0.3% worst-case relative error on *C. elegans* across all six targeted motif counts, and converge on every network in the set, while retaining only 13.9% edge overlap with the empirical connectome — against 11.6% for an unconstrained degree-preserving rewiring. The motif-matched surrogates are therefore not the connectome in disguise: they agree with it on the motif census and are otherwise as randomised as a standard null.

The held-out statistic confirms that the constraint is specific. Bi-fan counts, a 4-node motif deliberately excluded from the annealing objective, are not reproduced by the L3 ensemble, and — the point that matters — they are not even displaced in a consistent direction. In *C. elegans* the empirical count is $118,612$ against $179,732$ for the motif-matched ensemble, an overshoot; in the *Drosophila* central complex it is $824,107$ against $647,161$, an undershoot. Constraining the 3-node census therefore neither fixes nor systematically biases 4-node structure. The chain constrains exactly the structure whose contribution the ladder is designed to isolate and leaves higher-order structure free, which is what distinguishes L3 from a partial reconstruction of the connectome.

One further property is worth stating because the whole rung depends on it: the constraint has to be *binding*. If the empirical census were within ordinary sampling range of the unconstrained ensemble, L3 would be a relabelled L2 and the L2 → L3 contrast would be measuring nothing. It is not: the enrichment Z-scores of §3.1 are the distance between exactly these two ensembles' censuses, and they run from $Z = +2.7$ in the least-enriched net-

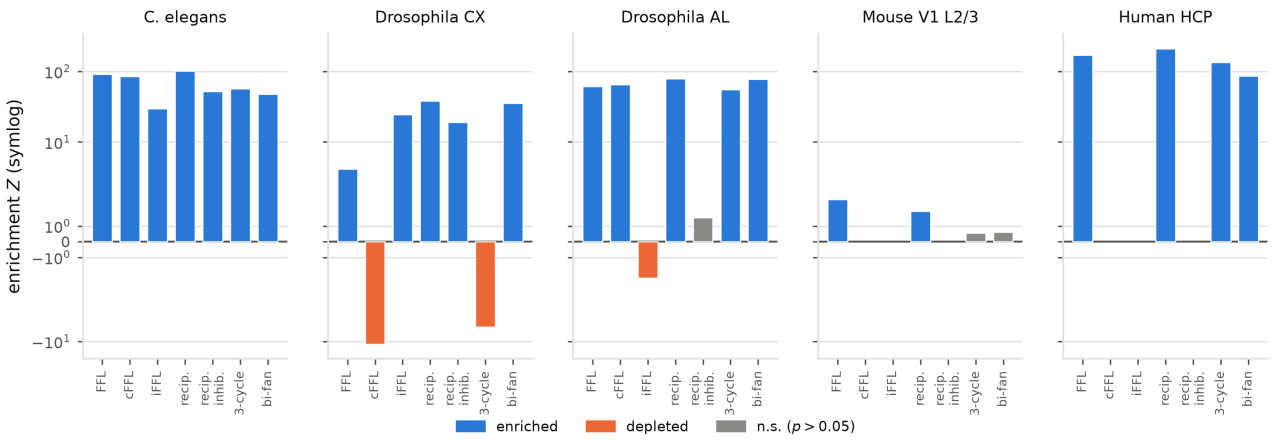


Figure 1: **Motif enrichment against a degree-preserving null**, $K = 500$ surrogates per network, exact counts, symlog axis. Colour encodes the outcome of each test rather than the network, because the point is that the outcome differs between networks: the *Drosophila* central complex *depletes* coherent feedforward loops and directed 3-cycles, the antennal lobe depletes incoherent feedforward loops, and mouse V1 L2/3 is close to its null throughout. Bars appear only where the dataset supports the motif's definition — coherent and incoherent feedforward loops require synaptic valence, which diffusion tractography does not carry and which the mouse release does not supply in connected form. The human column's magnitudes reflect an almost entirely reciprocal tractography graph rather than a stronger biological effect (see text).

work to $Z = +210.3$ in the most. Except in mouse V1, where the census is close to its null by construction and the contrast is correspondingly uninformative, L2 and L3 are separated by tens to hundreds of null standard deviations in the coordinates the chain constrains.

One asymmetry should be stated rather than left to the table. The L3 ensemble's members are somewhat more similar to *each other* than the unconstrained ensemble's members are (mean pairwise edge overlap 21.0% against 11.1% in *C. elegans*), which is expected — a constrained set is smaller than the set that contains it — but it means the L3 rung has less ensemble diversity than L2 at equal instance count. We report the pairwise overlap for every network so that this is visible, and we note that it works against detecting an L2–L3 difference only by inflating L3's variance, not by suppressing a mean difference.

3.3 Protocol sensitivity

Before comparing rungs we fix what the benchmark can detect. Under an all-to-all protocol, a reservoir with its recurrent matrix set to zero scores 59.6% on SHD while the *C. elegans* connectome scores 59.6%: the recurrent matrix is not used, and no structural comparison run in that regime can be informative. Under the sensory/motor-disjoint protocol used throughout this paper, the same zero-recurrence control falls to 4.9% against a chance level of 5.0%, while the connectome reaches 35.2%. Task-relevant signal must now traverse the recurrent graph, so differences between networks are attributable to wiring.

3.4 Decomposing performance across the ladder

Table 5 and Figure 2 give the decomposition. We read it property by property rather than as a count over the 20 contrasts, because the question is not how many differences reach significance but which structural properties this benchmark is sensitive to and in which direction.

The direction is the first thing to say, because it is uniform and it is not the direction the motif hypothesis predicts. *Every* structural property that measurably moves accuracy moves it *down*. Each rung adds something the connectome has and the rung below it does not, and at every step where the effect is resolvable the addition makes the network a worse reservoir.

Degree sequence (L0 → L1) is the largest single contributor: significant after Holm correction in 3 of 5 networks, median effect −4.65%, largest −8.02% (Hedges' $g = -2.38$).

The motif census (L2 → L3) is the contrast this paper was built to measure: it adds the connectome's exact 3-node motif spectrum while holding degree sequence and valence placement fixed, using the ensemble validated in §3.2. It is significant in 2 of 5 networks, with median effect −3.50% and largest −5.94% ($g = -2.06$). This is not a negligible effect and we do not report it as one — it is comparable in size to the degree-sequence contrast — but its sign is negative. Imposing a connectome's motif structure on a degree- and valence-matched surrogate costs accuracy on this task.

Valence placement (L1 → L2) — returning inhibitory labels to the neurons that carry them rather than permuting them across the graph — is significant in 1 of 5, median

Table 4: **Validation of the motif-preserving ensemble.** The chain must hit its target census, genuinely randomise, and not reproduce structure it was not asked to preserve. Columns 2–3: worst-case relative error across the six targeted motif counts, and the fraction of chains meeting the 2% convergence criterion. Columns 4–5: edge overlap with the empirical connectome for the motif-matched (L3) and degree-matched (L2) ensembles — L3 is close to the unconstrained level, so it is not the connectome in disguise. Columns 6–7: mean pairwise overlap *between* ensemble members, as evidence of mixing. L3 members are somewhat less mutually distinct than L2 members, which is expected because the constrained set is a subset of the unconstrained one; what matters is that they remain far from identical, so the chain explores the constraint surface rather than collapsing onto one region of it. This is empirical evidence of adequate mixing, not a proof of uniform sampling. Columns 8–10: bi-fan counts, a 4-node motif deliberately excluded from the objective, reproduced neither in value nor in a consistent direction, confirming the chain constrains 3-node structure specifically rather than reconstructing the graph.

	Target match		**Overlap w/ real (%)**		**Pairwise overlap (%)**		**Held-out bi-fan count**		
Network	**max err (%)**	**conv. (%)**	**L3**	**L2**	**L3**	**L2**	**empirical**	**L3**	**L2**
C. elegans	0.3	100	13.9	11.6	21.0	11.1	118,612	179,732	65,882
Drosophila CX	0.2	100	20.5	19.9	22.4	20.4	824,107	647,161	573,668
Drosophila AL	0.1	100	43.9	40.0	45.6	38.9	12,622,936	12,038,186	9,058,482
Mouse V1 L2/3	0.9	100	10.6	10.7	10.7	10.4	5,826	5,930	5,594
Human HCP	0.5	100	11.4	7.8	20.3	7.9	129,726	243,934	46,585

Table 5: **SHD test accuracy (%) by ladder rung**, mean ± 95% CI over 15 surrogates per rung. Every rung is matched on node count, edge count, spectral radius and input projection; adjacent rungs differ in exactly one structural property. Chance is 5.0%; a zero-recurrence network scores 4.9% under this protocol.

Rung	*C. elegans*	*Drosophila* CX	*Drosophila* AL	Mouse V1 L2/3	Human HCP
L0 density	37.1 ± 1.0	42.7 ± 0.6	39.9 ± 1.3	39.8 ± 1.0	38.8 ± 0.7
L1 +degree	36.8 ± 1.1	38.0 ± 1.3	31.9 ± 1.9	32.1 ± 1.9	39.1 ± 1.2
L2 +valence	36.9 ± 1.1	34.8 ± 1.5	30.6 ± 1.7	31.5 ± 1.9	38.3 ± 1.5
L3 +motifs	33.1 ± 1.4	33.9 ± 1.6	27.1 ± 2.7	31.8 ± 1.4	32.4 ± 1.4
L4 connectome	36.3 ± 1.7	26.3 ± 2.0	28.2 ± 1.6	31.8 ± 1.8	38.6 ± 1.0

−0.82%.

The residual wiring (L3 → L4) adds everything else about the empirical graph: weights aside, its exact synapse-by-synapse identity. Significant in 2 of 5, median 1.06%, and it is the one property whose sign is not consistent — it helps in some networks and hurts in others.

What the design could have detected. A small measured effect and an underpowered design look identical in a table, so we state the bound. With 15 surrogates per rung and 20 contrasts in the family, the design has 80% power to resolve a standardised effect of $d = 1.06$ uncorrected and $d = 1.54$ at the Holm worst case. The significant contrasts reported above sit at or beyond that bound; the non-significant ones do not exclude moderate effects, and we do not claim they do.

The reading. The motif hypothesis, as we stated it in §1 and as our own earlier work asserted it, is that connectome motif structure confers a computational advantage over matched controls. On this benchmark, in the fixed-reservoir regime, at matched size, density, spectral radius and input projection, it does not. The measured effect of adding the empirical motif census has the opposite sign, and it is large enough to detect in two of five networks.

We are careful about what this licenses, because the uniform negative sign across properties is itself informative about the probe. A reservoir with a linear decoder rewards distributed, high-dimensional state. Connectome structure — heavy-tailed degrees, excess feedforward loops and reciprocal pairs — concentrates and correlates activity, which spends the dimensions such a decoder needs. That is a property of this probe as much as of these graphs, and a different task with a different readout could order them differently. What it does establish is that an Erdős–Rényi comparison cannot be used to argue the motif hypothesis on this benchmark, because that comparison comes out against it.

3.5 The ladder at a matched operating point

One asymmetry in Table 5 needs separating out before any of it can be read as a statement about topology. The rungs are matched on spectral radius, but they do not fire at the same rate: averaged over networks the Erdős–Rényi rung settles at 10.8% mean rate against 6.1% for the empirical connectome, a factor of 1.77. Across all runs firing rate

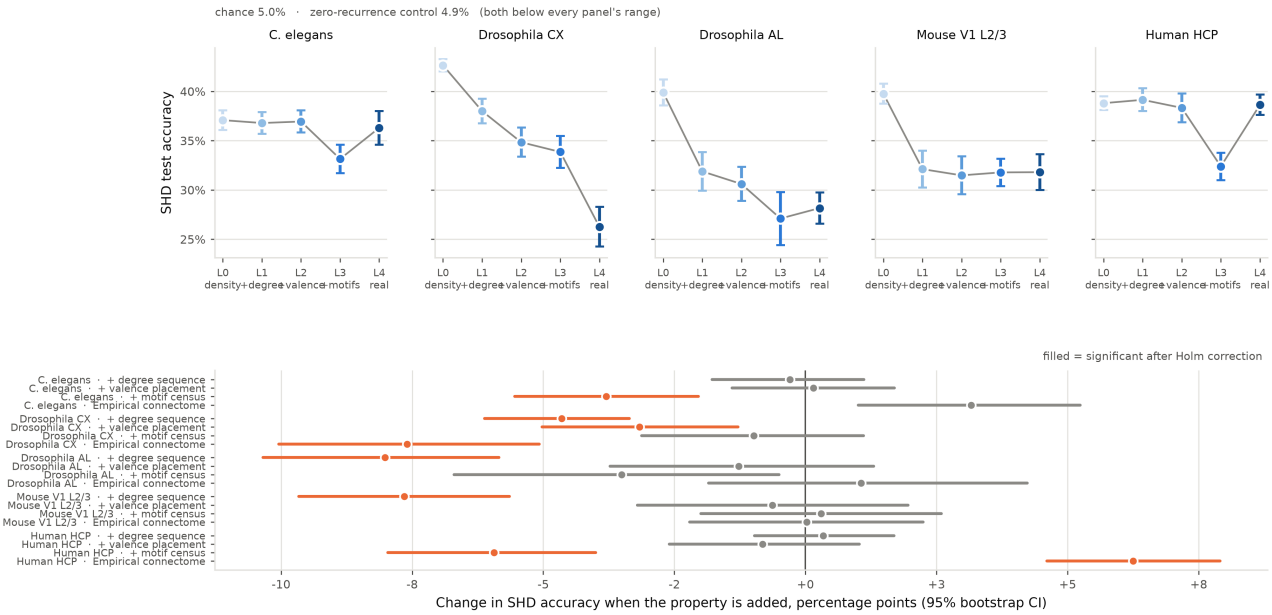


Figure 2: **Structural decomposition of reservoir performance on SHD. Top:** mean accuracy with 95% CIs at each ladder rung, per network. Each rung adds one structural property to the rung on its left; all rungs are matched on node count, edge count, spectral radius and input projection. Dotted line, chance; dashed line, the zero-recurrence control. **Bottom:** the contribution of each property, as the difference between adjacent rungs with bootstrap 95% CIs; intervals spanning zero indicate a property that does not measurably change performance.

Table 6: **Contribution of each structural property**, as the difference between adjacent ladder rungs in SHD accuracy (percentage points), with percentile-bootstrap 95% CIs, Hedges' g, and Holm-corrected p-values across the whole family of contrasts. $*$ marks $p_{\text{Holm}} < 0.05$.

Network	Property added	Δ (pp)	95% CI	g	p_{Holm}
C. elegans	+ degree sequence	-0.29	[-1.78, +1.11]	-0.14	1.000
C. elegans	+ valence placement	+0.15	[-1.40, +1.70]	+0.07	1.000
C. elegans	+ motif census	-3.80	[-5.54, -2.04]	-1.46	0.005∗
C. elegans	Empirical connectome	+3.16	[+1.01, +5.24]	+0.99	0.115
Drosophila CX	+ degree sequence	-4.65	[-6.12, -3.36]	-2.31	0.000∗
Drosophila CX	+ valence placement	-3.16	[-5.02, -1.29]	-1.14	0.045∗
Drosophila CX	+ motif census	-0.98	[-3.13, +1.11]	-0.31	1.000
Drosophila CX	Empirical connectome	-7.60	[-10.05, -5.07]	-2.04	0.000∗
Drosophila AL	+ degree sequence	-8.02	[-10.35, -5.85]	-2.38	0.000∗
Drosophila AL	+ valence placement	-1.28	[-3.72, +1.30]	-0.34	1.000
Drosophila AL	+ motif census	-3.50	[-6.70, -0.50]	-0.76	0.465
Drosophila AL	Empirical connectome	+1.06	[-1.85, +4.23]	+0.24	1.000
Mouse V1 L2/3	+ degree sequence	-7.65	[-9.66, -5.65]	-2.51	0.000∗
Mouse V1 L2/3	+ valence placement	-0.63	[-3.21, +1.96]	-0.16	1.000
Mouse V1 L2/3	+ motif census	+0.30	[-1.99, +2.59]	+0.09	1.000
Mouse V1 L2/3	Empirical connectome	+0.02	[-2.21, +2.24]	+0.01	1.000
Human HCP	+ degree sequence	+0.34	[-0.97, +1.69]	+0.18	1.000
Human HCP	+ valence placement	-0.82	[-2.59, +1.02]	-0.31	1.000
Human HCP	+ motif census	-5.94	[-7.96, -4.00]	-2.06	0.000∗
Human HCP	Empirical connectome	+6.25	[+4.61, +7.90]	+2.52	0.000∗

is correlated with accuracy ($r = 0.38$), and it determines the synaptic-operation count outright. A comparison at matched gain therefore confounds topology with operating point, in the direction that flatters the unstructured control.

Recalibrating each network individually to a common mean rate of 0.07 removes the confound, at the cost of letting spectral radius vary. Table 7 and Figure 3 give the result, and it is not the one we expected.

Where the calibration fails, and why that is a result. Of 250 calibrated surrogates, 27 never reach the target rate at all: the bisection saturates at the top of its bracket and the network still fires below target, in the worst case at 49% of it. 25 of the 27 are in one network, *DrosophilaCX*, whose edges are 66% inhibitory — raising the gain in such a circuit raises its inhibition too, so its attainable firing rate is bounded above by its own valence organisation. The effect strengthens down the rungs, from none at L0 and L1 to every surrogate at L4, which places the ceiling squarely on the empirical valence placement rather than on density. We exclude those cells from every matched-rate statistic rather than averaging them in, since a cell that was not calibrated is not rate-matched, and we report the exclusion because the ceiling is a property of the network worth stating. That leaves 2 networks in which every rung is fully matched, and the contrasts below are computed on those.

The operating point does not explain the gap. At a matched operating point the Erdős–Rényi rung still outperforms the empirical connectome, by 9.00% to 13.95% of accuracy in percentage points, against 7.96% to 11.74% for the same networks at matched gain. The gap does not shrink; it is marginally larger. Firing rate is therefore a real confound in the matched-gain comparison — it is why we report both — but it is not what produces the unstructured control's advantage, and we were wrong to expect that it would.

What the two versions together establish is narrower

Table 7: **The ladder at a matched operating point.** Each network is calibrated individually, by bisection on its own spectral radius, to a common mean firing rate of 0.07; 10 surrogates per rung. **Top:** SHD test accuracy (%), mean ± 95% CI, over the surrogates that reached the target. **Bottom:** the spectral radius each topology required, averaged over all surrogates including those that did not. 27 of 250 surrogates never reached the target rate: the bisection saturates at the top of its bracket and the network still fires below target. * marks a cell where some surrogates were excluded on that ground and † one where all were — the *Drosophila* central complex is 66% inhibitory, so raising its gain raises its inhibition and its attainable rate is capped. Accuracy differences that survive in Table 5 but not here were differences in activity level rather than in wiring.

Rung	*C. elegans*	*Drosophila* CX	*Drosophila* AL	Mouse V1 L2/3	Human HCP
Test accuracy (%)					
L0 density	39.0 ± 1.2	41.9 ± 0.6	42.0 ± 0.8	41.0 ± 0.9	41.0 ± 1.2
L1 +degree	37.1 ± 1.6	37.9 ± 1.3	32.7 ± 2.8	32.8 ± 1.9	38.2 ± 0.8
L2 +valence	36.1 ± 1.3	$35.1 \pm 1.9^*$	31.0 ± 1.7	32.4 ± 1.9	38.6 ± 1.1
L3 +motifs	$35.0 \pm 1.4^*$	$39.0 \pm 5.0^*$	28.6 ± 2.2	32.8 ± 1.8	$34.3 \pm 1.4^*$
L4 connectome	35.9 ± 1.9	†	28.1 ± 1.9	32.0 ± 2.3	38.9 ± 0.8
Calibrated spectral radius ρ					
L0 density	1.22	1.15	1.12	1.15	1.17
L1 +degree	1.28	1.62	1.26	1.25	1.28
L2 +valence	1.26	5.96	1.36	1.23	1.28
L3 +motifs	1.41	5.79	1.38	1.21	1.46
L4 connectome	1.30	6.00	1.35	1.25	1.31

and firmer than either alone. On this task, in this regime, connectome wiring is not merely no better than an unstructured graph of the same size and density: it is measurably worse, and that remains true when the networks are compared at the same activity level. Reading down the rungs shows where the cost is incurred. The step that imposes the empirical degree sequence, L0 → L1, accounts for 67% to 91% of the whole decline; the three steps that follow — adding valence placement, the motif census and the remaining wiring — divide the rest between them and are individually small. Degree heterogeneity, not motif structure, is the structural property this benchmark is sensitive to — and its effect has the opposite sign to the one the motif literature would predict for connectome structure as a whole.

We are explicit about what this does not show. It does not show that connectome wiring is bad for computation, but that a degree-heterogeneous graph is a poor fixed reservoir for a 20-class classification with a linear readout, which is one task in one regime and not the objective any nervous system is under selection for. It does show that a comparison against an Erdős–Rényi control cannot be used to argue that connectome structure confers a computational advantage, because on this benchmark that comparison comes out the other way.

3.6 Which rewiring counts as the null

This section and the three that follow it vary one argument of Eq. 1 at a time, holding the rest fixed, and report what that argument alone is worth. We begin with $\mathcal{N}$, because it is the argument whose specification is least determinate and, as it turns out, the one worth the most.

Every comparison in this literature is a comparison against a rewiring, and the rewiring is almost always specified in a phrase: *degree-preserving randomisation*, or *randomised while preserving density and degree sequence*. That phrase does not determine an ensemble. At least two constructions satisfy it, and they differ in what happens to the weights:

- **Weight-carrying rewiring.** Double-edge swaps move edges, and each weight travels with the edge that carries it. The Brain Connectivity Toolbox routine `randmio_und_connected` / `randmio_dir_connected` [24, 25] does this, additionally rejecting swaps that disconnect the graph. This is what `conn2res` calls, and therefore what the published result was measured against.

- **Rank-reassigned rewiring.** The binary topology is randomised by the same swaps, and the original multiset of weight values is then re-assigned to the new edges in order of endpoint-degree product. This preserves the degree sequence and the weight *distribution* exactly, and destroys the association between a weight and the edge carrying it.

Both are defensible nulls and both answer to the published phrase. We wrote the second one, believing we had written the first, and the error is instructive enough that we ran both rather than deleting it: it preserves every marginal the description names, it produced a clean and

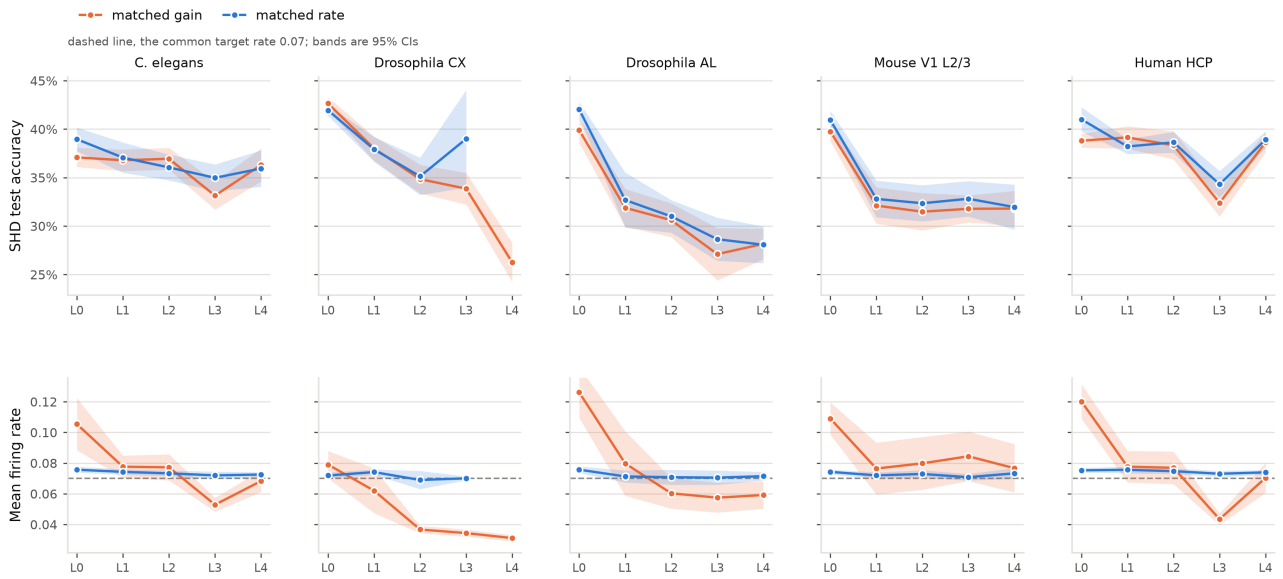


Figure 3: **The ladder at matched gain and at a matched operating point.** Colour encodes the matching regime rather than the network, because the comparison is within a network. Rungs are L0 density, L1 +degree, L2 +valence, L3 +motifs, L4 the empirical connectome. **Top:** SHD test accuracy by rung under each regime. **Bottom:** mean firing rate. At a fixed spectral radius (orange) the rungs run at different rates, and the accuracy ordering tracks the rate; calibrating each network individually to the common rate (blue, dashed line) holds every network on the target. The accuracy decline from L0 to L4 survives that correction. Cells whose calibration never reached the target are omitted from the matched-rate series, which is why the *Drosophila* CX panel has no L4 point. Bands are 95% CIs.

highly significant result, and nothing about its output indicated that it was answering a different question. It was caught by reading the toolbox source, not by anything in the numbers.

The two ensembles do not agree. Figure 4A shows the result and Table 8 the full design against both. Of 10 (network, unit model) cells, the two nulls give the connectome's advantage *opposite signs* in 6, and opposite *significant* verdicts — one says the connectome leads, the other says it trails, both at $p < 0.05$ — in 3. The median absolute disagreement between the advantage measured against one null and against the other is 0.73 capacity units, and the largest is 6.98. For scale, these are differences of the same order as the effects the literature reports.

Which one reproduces the published direction. Against the weight-carrying ensemble — the one the original toolbox builds — the graded-rate reservoir puts the connectome ahead in 1 of 5 networks and behind in 3, median −0.51. Against the rank-reassigned ensemble the same graphs, the same unit model and the same operating point put it ahead in 5 of 5, median +2.13. The direction of the headline result in this design is set by a choice that the method sentence does not record.

Why they differ, and a prediction that could have refuted it. The disagreement is not noise, and it has a structural cause specific enough to be checked in advance (Fig. 4B,C). Sorting the weight multiset onto rewired edges by endpoint-degree product does not merely randomise: it *installs* an association between an edge's weight and its endpoints' degrees. Across the 4 weighted networks, carrying weights with their edges shifts the weight–degree correlation by a median of +0.06 relative to the empirical graph, and never by more than 0.16; rank-reassignment shifts it by +0.66. In mouse the empirical association is slightly *negative* (−0.10), `randmio` returns +0.06, and rank-reassignment returns +0.49; in rat the corresponding figures are +0.09, +0.01 and +0.82. A null that adds a property the data lacks is not a control for the absence of that property, and a reservoir whose largest weights are concentrated on hub-to-hub edges is a poor one — so the connectome's advantage over it is inflated, in the direction and roughly the magnitude we observed.

This account makes a prediction that would have falsified it cheaply: if the difference comes from permuting weights, then on a network with no weight variation the two constructions are the same ensemble and must agree exactly. One network in the set is binary. On macaque the two nulls shift the weight–degree association by −0.00 and −0.00 respectively — that is, by nothing — and the connectome's measured advantage against them differs by 0.008 capacity units, against a median disagreement of 0.73 elsewhere. It is the one cell in Table 8 where the choice of null provably cannot matter, and it is the one

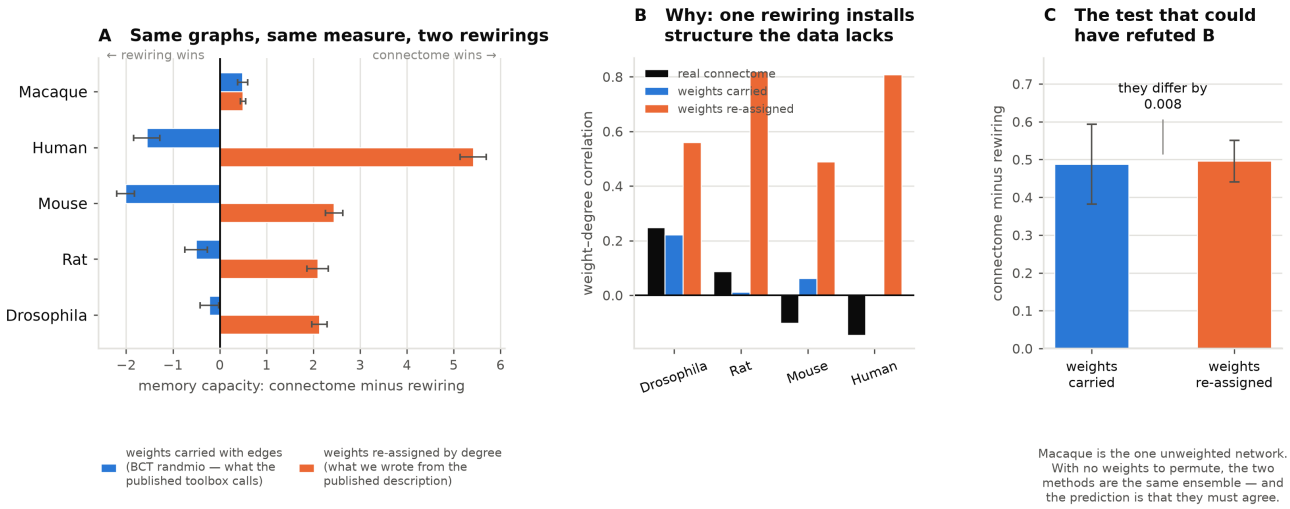


Figure 4: **Two rewirings that answer to the same description, and the answers they give. (A)** The empirical connectome's memory-capacity advantage over each of two degree-preserving rewirings, at $\rho = 1$, with graded units. Same graphs, same inputs, same delays, same readout, ten paired instances, 95% confidence intervals. The only thing that differs between a blue bar and the orange bar beside it is what happened to the edge weights during randomisation: carried along with their edges, or re-assigned to the rewired graph by endpoint-degree rank. Both procedures preserve the degree sequence and the weight distribution exactly, and both are described in the literature as "degree-preserving randomisation". In four of five networks the pair points in opposite directions. **(B)** The reason. Rank-reassignment does not only randomise: it installs an association between an edge's weight and its endpoints' degrees that the empirical graphs do not have (black) and that weight-carrying rewiring does not create (blue). A reservoir whose largest weights sit on hub-to-hub edges is a poor one, so the connectome's measured advantage over it is inflated. **(C)** The observation that makes (B) an explanation rather than a story, because it could have come out the other way. If the difference arises from permuting weights, then on a network with no weight variation the two procedures are the same ensemble and must agree. Macaque is unweighted. There the two measured advantages differ by 0.008 capacity units, against a median disagreement of 0.73 across the rest of the set.

cell where it does not.

What this does and does not establish. It does not show that either null is wrong. Rank-reassignment is a coherent ensemble and a comparison against it is a coherent question — it asks whether the *placement* of a connectome's weights matters given its degree sequence, which is a narrower and in some ways sharper question than the one weight-carrying rewiring asks. What it shows is that the two questions have different answers, that the phrase in general use does not distinguish them, and that a reader cannot recover which was asked from the reported number. A replication that constructs its own null from the published description is therefore not a replication, and we include ours as an instance of that failure rather than a warning about it.

3.7 What a firing threshold does to the comparison

The disagreement of §3.8 has a candidate explanation that does not involve the operating point at all, and it is checkable in one experiment. [9] evaluate an echo state network of tanh units. We evaluate a spiking leaky integrate-and-fire network. Those are not two implementations of one model. A tanh unit responds to any input however small and is never silent; a LIF unit below threshold emits nothing and contributes no variance to the readout at all.

We therefore ran both unit models over identical graphs, identical rewirings, identical inputs, identical delays and an identical readout, on the five regional connectomes distributed with `conn2res` — the networks the published claim is made on, so that the comparison inherits neither our subgraph extraction nor our binarisation. 2700 evaluations: 5 networks × 10 paired instances × 9 gains × two unit models.

All numbers in this subsection are measured against `randmio`, the ensemble the published toolbox builds, for the reasons given in §3.6.

With graded units, against the published null, we do not recover the published direction. The empirical connectome leads its rewiring in 1 of 5 networks at $\rho = 1$ and *trails* significantly in 3, median −0.51 capac-

Table 8: **The connectome advantage depends on which rewiring it is compared against, and on what a unit is.** Empirical connectome against two degree-preserving rewiring ensembles at $\rho = 1$, the operating point of the published comparison, on the five regional connectomes distributed with `conn2res`. `randmio` carries each weight along with its edge; rank-reassignment rewires the topology and then re-assigns the same multiset of weights by endpoint-degree rank. Both satisfy "degree-preserving randomisation". Identical graphs, inputs, delays and readout; ten paired instances; paired t-test, $*$ marks $p < 0.05$. Absolute capacities are shown because they establish whether a cell is informative: the human spiking row sits at the floor and carries nothing, the other four do not.

	Connectome	BCT `randmio`		Rank-reassigned	
Network	**Capacity**	**Capacity**	**Difference**	**Capacity**	**Difference**
Graded-rate units (`tanh`)					
Human	11.46	13.02	$-1.56 \pm 0.28^*$	6.04	$+5.42 \pm 0.28^*$
Macaque	8.67	8.19	$+0.49 \pm 0.11^*$	8.18	$+0.50 \pm 0.06^*$
Mouse	7.19	9.20	$-2.01 \pm 0.19^*$	4.75	$+2.44 \pm 0.19^*$
Rat	7.46	7.97	$-0.51 \pm 0.24^*$	5.37	$+2.09 \pm 0.22^*$
Drosophila	5.92	6.14	-0.22 ± 0.20	3.79	$+2.13 \pm 0.16^*$
Spiking units (LIF)					
Human	0.05	1.36	-1.31 ± 1.29	0.02	$+0.03 \pm 0.05$
Macaque	1.93	1.88	$+0.04 \pm 0.07$	1.94	-0.01 ± 0.04
Mouse	1.34	1.61	$-0.27 \pm 0.08^*$	1.56	$-0.23 \pm 0.11^*$
Rat	1.22	1.20	$+0.02 \pm 0.05$	1.16	$+0.06 \pm 0.07$
Drosophila	1.09	1.09	$+0.00 \pm 0.01$	0.97	$+0.12 \pm 0.25$

ity units. Against the rank-reassigned ensemble, on the same graphs at the same operating point in the same unit model, it leads in 5 of 5, median +2.13.

We do not read this as a refutation of [9]. Our harness differs from theirs in the input projection, the delay set and the ridge solver as well as the null, and any of those could account for a reproduction failure. What it does establish is narrower and, we think, more useful: the choice of null *alone* moves the result by more than the effect being reported, which makes it a sufficient explanation for a disagreement of this size without anything else having to be wrong. A reproduction attempt that differs from the original in its null is not evidence against the original.

With spiking units the comparison is weaker still. On the same graphs the connectome leads significantly in 0 of 5 networks at $\rho = 1$ and trails significantly in 1; the median difference is +0.00. Across the gain axis it leads in 2% of cells. Capacity itself falls by a factor of 4.9, from 7.0 to 1.4, so part of the weakening is simply that there is less signal to resolve — and the next paragraph asks whether the rest is a fact about spiking units or about our choice of constants.

Is the spiking model simply untuned? The obvious objection is that we ran the spiking reservoir at one threshold, one leak and one input scale, none of them tuned, and that a quiet network is a misconfigured one rather than an informative one. That objection deserves an experiment rather than a paragraph, and it gets one designed to refute us: we swept input gain from 0.5 to 16, threshold from 0.25 to 1 and two leaks — 36 configurations, 1728 evaluations — and asked whether the connectome advantage appears *anywhere* the network is computing.

It does, in a few places, and not enough of them to sustain the strong claim. Of 144 (network, configuration) cells, 80 are healthy by the pre-stated criterion that the rewired control retains capacity above 0.5 with fewer than 80% of units silent. Among those, the connectome leads significantly in 3 and *trails* significantly in 7, with a median difference of -0.16. The configuration used in Table 8 is not a pathological corner: it sits within 20% of the best capacity the sweep found.

So the accurate statement is not that the advantage is absent under spiking units but that it is *sporadic and inconsistent in sign* — present in 3 of 80 well-conditioned settings, reversed in more of them, and negative at the median. That is a weaker claim than the one we set out to make, and it is weaker in the same direction as everything else in this section: once the null is the published one, the unit model is no longer carrying a clean reversal, because there is no longer a clean effect in either model for it to reverse.

The silencing mechanism, which we had previously reported as refuted. We predicted the mechanism before running it: a heavy-tailed degree sequence should

leave a connectome's low-degree neurons below threshold, silencing them, while no tanh unit can be silent. The silencing is real and large — 60.7% of spiking units never fire, against 0.6% of rate units. Whether it is also *differential* is a question the choice of null decides, and we have had both answers. Against the rank-reassigned ensemble the silencing gap did not predict the capacity gap and we reported the prediction as failed. Against `randmio` it does: across 45 (network, gain) cells, the amount by which a connectome silences more than its rewiring predicts the amount by which it loses capacity, $r = -0.58$, $p = 2.5 \times 10^{-5}$.

We report the sign change rather than only the surviving result, because the two conclusions were drawn from the same code and the same graphs and differ only in the control. The supported version is the one measured against the published ensemble, and it is a modest claim: differential silencing accounts for part of the spiking arm's behaviour, in the predicted direction, leaving most of the variance unexplained. Thresholding also compresses the measurement for every topology alike — mean capacity falls from 7.0 to 1.4, a factor of 4.9 — and that compression is not differential at all.

A floor, declared. For the human network the spiking capacities are 0.049 against 0.019 — both effectively zero, so that network is uninformative in the spiking arm whatever it shows. The remaining four retain capacities between 1.0 and 1.9 and do carry information. The claim rests on those four.

What this licenses. Less than we hoped, and something worth saying anyway. We cannot claim a clean reversal between unit models, because against the published null there is no clean effect in the graded model to reverse. What we can claim is that the two model classes do not give the same answer on identical graphs, that capacity in the spiking model is compressed by a factor of 4.9 so that effects of the reported size are near the resolution of the measurement there, and that part of the difference tracks differential silencing. The practical consequence is unchanged by the weakening: the connectome-reservoir result is routinely invoked as motivation for neuromorphic architectures, neuromorphic hardware is spiking, and the evidence that the property survives the transfer does not exist in either direction yet.

3.8 The contrast changes sign along the gain axis

Everything reported so far was measured at one operating point. §3.5 showed that the choice of operating point is not innocuous even when spectral radius is held fixed. The question this raises is sharper than a robustness check, because the literature reaches the opposite conclusion to ours on closely related measurements, and does so at a different point on the same axis.

Suárez et al. [9] report that empirical connectomes have significantly greater linear memory capacity than degree-preserving rewirings, and report it *at criticality*, at a spectral radius of 1. Our ladder is evaluated at $\rho = 1.30$ and finds the opposite, on that same probe (§3.10). Both measurements can be right, and we show that they are: the contrast is not a fixed property of the pair of graphs being compared. It changes sign along the gain axis.

Sweeping the axis. We re-ran the full ladder across 7 spectral radii from 0.80 to 2.50, 8 surrogates per rung per gain, each surrogate swept across the whole axis so that curvature in ρ is not confounded with surrogate-to-surrogate variation. Of 1400 evaluations 2 failed and are excluded and counted rather than averaged in.

Two things follow, and the first is not what we expected to find.

On task accuracy the published direction does not appear, at any operating point we tested including criticality. At $\rho = 0.95$ the Erdős–Rényi rung leads the empirical connectome in 5 of 5 networks, with every one of those differences significant against its own paired instance variation; the median gap is 7.33%. At $\rho = 1.30$ it is 7.21%. Whatever advantage connectome topology has at criticality, it is not visible in the accuracy of a linear readout on this task.

This is consistent with what has recently been reported from a hyperparameter sweep of the *C. elegans* connectome against Erdős–Rényi, Watts–Strogatz and shuffled controls [33], which finds the connectome in a lower-peak-performance, lower-sensitivity regime and notes that the connectome-versus-null contrast varies with raw spectral radius. Our contribution on this point is not the variation, which that work identifies, but its extent.

The contrast is not merely gain-dependent; it is not a constant of the pair being compared. It changes sign along the axis in 3 of 5 networks, with the contrast significant on both sides of the crossing in each case, and the largest within-network swing is 23.39% of accuracy — several times the size of any structural effect in §3.4. In those networks the connectome catches the unstructured rung and overtakes it above $\rho \approx 1.3$. In the remaining two the unstructured rung leads significantly at every gain on the grid. We note the disagreement plainly: [33] reports the connectome-versus-null relationship as consistent across the regimes it tests, and we find a reversal. The two are not straightforwardly comparable — that work sweeps one connectome on task-agnostic probes, ours sweeps five on task accuracy, and our own probe-based sweep is underpowered (below) — so we record this as a discrepancy to be resolved rather than as a correction. The direction of the comparison is therefore a joint property of the topology *and* the regime, and a single number reported for it is a number about both.

Attempting the published comparison directly, and failing to resolve it. The bottom panel of Fig-

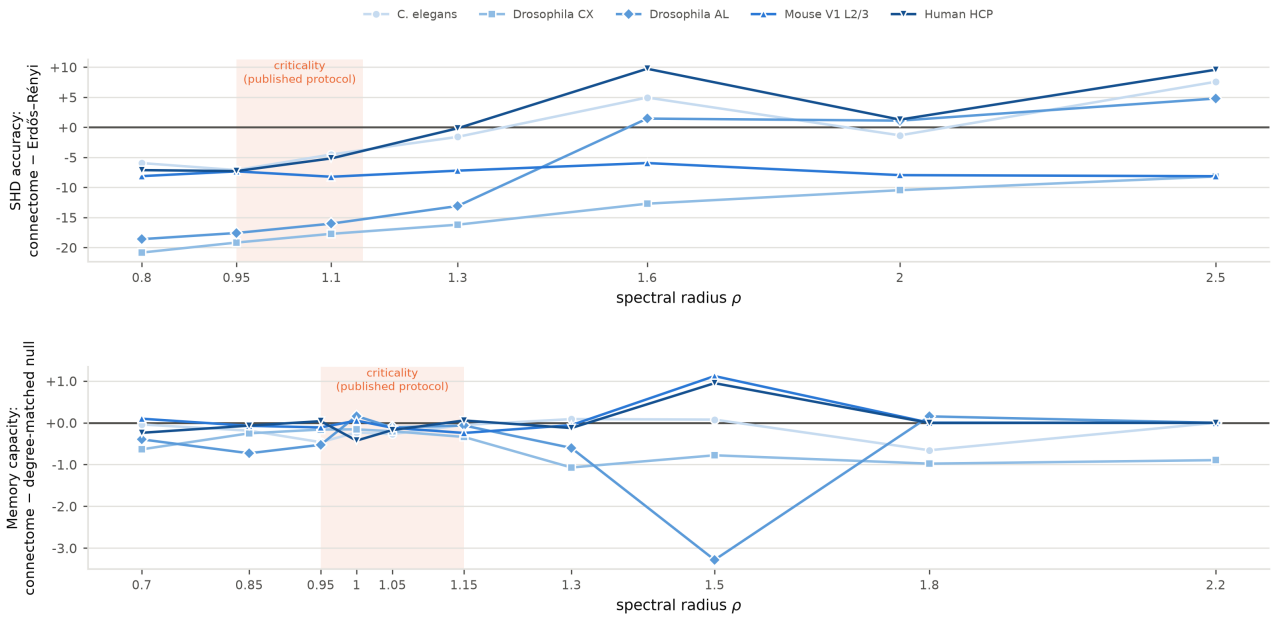


Figure 5: **Where the comparison is made decides what it finds.** Both panels plot a *contrast*, not a performance curve, so that zero is the line that matters: below it the connectome leads, above it the null does. **Top:** SHD accuracy, empirical connectome minus the Erdős–Rényi rung. **Bottom:** linear memory capacity, empirical connectome minus the degree-preserving null — the quantity and the control used by [9]. The shaded band marks the critical region ($\rho \approx 1$) at which that comparison is reported. A published result of this kind is a single point on one of these curves.

ure 5 repeats the measurement the published claim rests on: linear memory capacity, empirical connectome against a degree-preserving null, across 10 gains from 0.70 to 2.20, ten surrogates per rung per gain, paired by instance.

We do not detect the reported advantage. At $\rho = 1.00$ — the published operating point — the connectome leads in 2 of 5 networks by the sign of the mean, and in 0 of 5 once the paired confidence interval is taken into account. The median difference is -0.157 capacity units, against the null.

This is a non-detection and not a refutation, and the distinction is the whole of it. Our design at this operating point resolves a median effect of 1.04 capacity units at 80% power — 46% of the null network's own capacity. An advantage of the size that would be scientifically interesting here is comfortably inside our confidence intervals. On 1 of 5 networks the variance is low enough to resolve 8% of capacity, and there the difference is $+0.05 \pm 0.11$; on the other four the attempt is uninformative, and we say so rather than reporting four null results as though they were evidence.

Several differences from the original could account for a real effect going undetected here, and we have not isolated which. Our networks are size-matched induced subgraphs of 400 nodes rather than whole released connectomes; our spiking reservoir is a binary-weight LIF network rather than a rate model; and our memory-capacity estimator, input scaling and washout are our own. We report the attempt because a reader deciding what to believe about the published result should know that an independent implementation did not see it, and should equally know that this implementation could not have seen it unless it were very large.

Against an unstructured graph the direction is consistent. One asymmetry survives the power problem, because it is large. The published comparison is against a *degree-preserving* null. Against the Erdős–Rényi rung the connectome's memory capacity is lower at essentially every gain and in every network, by margins that reach several capacity units — far outside the intervals above. Whatever the degree sequence and the wiring buy relative to each other, they do not recover what is lost relative to an unstructured graph of the same size and density. This is the same ordering the accuracy sweep gives, and the same one the ladder gives at a fixed gain.

Where the sweep stops being a comparison. A gain axis has an end, and past it the question dissolves. As ρ grows the reservoir saturates: every neuron fires at every step, the readout features stop varying, and the state carries no information about the input regardless of how it is wired. We record the effective rank of the readout features for every cell as a fraction of their number, so that regime is identifiable by a number rather than inferred from a surprising accuracy. It falls from 88% at $\rho = 1.30$ to 73% at the top of the grid. Of 1400 evaluations, 2 failed outright and are excluded and counted rather than averaged in as missing values.

We therefore do not rest the sign change on the saturated end of the axis. The crossing reported above occurs between the critical region and $\rho = 1.30$, where the state is still high-rank in every network, and the far end of the sweep is shown for completeness rather than used as evidence.

Two protocols, two questions. Neither way of choosing an operating point is wrong, and the difference between them is not a matter of care. Comparing each network at its own optimum asks how good a topology can be made to look when the rest of the model is tuned for it; comparing at a common point asks what the topology does to a network held otherwise fixed. Under the first protocol the rungs peak at different gains — median 1.10 for the Erdős–Rényi rung against 2.00 for the connectome — and the contrast at each rung's own best gain is 4.33%, negative in 0 of 5 networks.

We draw a methodological conclusion rather than a biological one. A single-operating-point comparison between a connectome and a null establishes a fact about that operating point. Reported without the curve, it is indistinguishable from a fact about the topology, and the two differ by a sign over a range of ρ that any two groups might reasonably choose differently. This applies to our own matched-gain and matched-rate results in §3.4 and §3.5 exactly as it applies to the literature's, and it is the reason we report the axis rather than a point.

3.9 The null result is not an artefact of the sampler

Because L3 is produced by an annealed chain rather than by a construction with a proof behind it, two objections have to be answered before a null L2–L3 contrast can be read as a statement about motifs. Neither is addressed by the ensemble validation of §3.2, which establishes only that the chain hits its target, randomises, and does not reproduce held-out structure.

The first is mixing. If the chain has not moved far enough, its surrogates retain memory of where they started and a null contrast means only that the chain was too short. We therefore re-ran the L3 rung over chain lengths from 30 to 240 sweeps, a factor of eight around the value used throughout. Mean accuracy moves by a median of 1.04% across that range, and at most 3.66% on any network (*humanHCP*). The reference to judge that against is the surrogate-to-surrogate standard deviation at a *single* chain length, which has a median of 3.40%: the variation attributable to an eightfold change in chain length is of the same size as the variation between surrogates drawn at one setting. It is also not a drift — accuracy is non-monotonic in chain length in 4 of 5 networks, which is the signature of a chain that has stopped moving rather than one still travelling. Both are far below the degree-sequence effect the ladder resolves. We read this as showing the result is not chain-length limited, not as showing the chain has provably mixed.

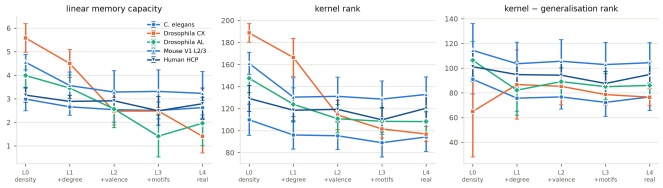


Figure 6: **Dynamical probes across the ladder.** Memory capacity, kernel rank and the capability index (kernel − generalisation rank) as a function of ladder rung. The probes are independent of the benchmark, and they reproduce its ordering: the unstructured rung leads, and the decline is concentrated in the early rungs.

The second is the starting point. The chain we use begins at the empirical connectome and walks under the census constraint; the opposite construction begins at a fully rewired graph and anneals towards the census. The two guarantee opposite things, and where both reach the target they should agree. Of the 5 networks, the random start converges on 4; restricted to those, the two constructions differ in mean accuracy by at most 1.07%. Where the random start fails it fails by not reaching the census at all, which is the reason the empirical start is used in the main text rather than a preference between two equally good options.

3.10 The same decomposition on intrinsic dynamics

A result read off a task metric invites the objection that the readout, not the reservoir, is what is being measured. We therefore repeated the decomposition on three task-independent probes of the reservoir itself — linear memory capacity, kernel rank and generalisation rank — which are the standard measures of a reservoir's computational capability and do not involve the benchmark at all. These were run at 10 surrogates per rung rather than 15, so Holm correction across the same 20-contrast family is correspondingly harsher and the significance counts are not comparable with those of §3.4; the effect sizes and the signs are.

The probes reproduce the task result, including the part of it we did not expect (Figure 6). The Erdős–Rényi rung has the higher memory capacity in 5 of 5 networks, by a median factor of 1.41, and the higher kernel rank in 5 of 5, by 1.21. So the connectomes are not trading task accuracy for some capability the benchmark fails to reward; they have less of the capability the probes measure.

The ordering of properties also carries over. Imposing the degree sequence lowers memory capacity in 5 of 5 networks (median Hedges' $g = -0.58$) and kernel rank in 5 of 5 ($g = -0.97$), while the motif census moves them least of any rung ($g = -0.05$ and -0.29). After Holm correction 1 of 20 contrasts survive in memory capacity and 1 of 20 in kernel rank — in both cases the same one, valence

placement in the *Drosophila* central complex, the network whose edges are two-thirds inhibitory.

The two classes of measurement agree on the direction — every resolvable effect is negative, and the unstructured rung leads — and they agree that the degree sequence carries the largest share. They do *not* agree on the motif rung. On task accuracy the motif census produces an effect comparable to the degree sequence's (-3.50% against -4.65%); on the dynamical probes it is the smallest contrast in the ladder. We do not have an account of that divergence and do not want to manufacture one. The most likely reading is that the probes measure properties of the state ensemble that the ridge readout does not fully exploit, so a structural change can cost task accuracy without costing measured capacity — but that is a conjecture, and the honest summary is that the benchmark and the probes concur on the headline ordering and part company on one rung of it.

3.11 Degeneracy is a property of directions

The measured spectral gaps of these networks are as small as 0, so the Davis–Kahan bound admits perturbations of norm at most 2.2×10^{-2} — far below any perturbation that a single synapse removal produces. The classical bound is not loose here; it is inapplicable, and the ratio between empirical tolerance and that bound, which an earlier revision of this work reported as evidence of biological robustness, is a property of eigenvector conditioning.

Proposition 2 instead predicts that tolerance scales inversely with the operational alignment γ of the perturbation direction. We tested that in two forms, and they do not give the same answer.

The ordinal prediction holds, mostly. Within each network, γ is larger for the structured directions than for the random one: relative to uniformly random removal, degree-targeted removal realises $2.6\times$ the worst case and the first-order maximally damaging ranking $3.0\times$. The tolerated fractions run in the opposite order, as predicted — 83% for random removal, 73% for degree-targeted and 64% for adversarial, a factor of ≥ 1.3 between the extremes at identical cardinality (Figure 7). Taken network by network, γ orders the three strategies correctly in 3 of 5.

The quantitative prediction does not. The proposition says more than an ordering: it says the tolerated magnitude scales as $1/\gamma$, so the product $\gamma \cdot \|E\|_F^{\max}$ should be constant across directions within a network. It is not. That product still varies by a median factor of 2.2 between strategies — a reduction on the $3.0\times$ spread in γ alone, but not the collapse the proposition predicts. We report this as a partial failure of our own prediction rather than restricting the claim to the ordering after the fact: γ identifies which direction is more damaging, and does not correctly predict by how much.

Table 9: **Logic synthesis of the accompanying RTL** with Yosys and ABC onto the Nangate45 Open Cell Library (typical corner). These are the quantities this flow can establish. No die area, power or energy-per-operation figure is reported: the flow performs no placement, routing, clock-tree synthesis or parasitic extraction. No clock frequency is reported either, because the toolchain ships no static timing analyser. The commercial-process figures reported in earlier revisions of this project were never measured and are withdrawn.

Module	Cells	Flip-flops	Cell area (μm^2)
`dendritic_lif_core`	4,401	70	5,301
`synaptic_accumulator`	19,649	8,192	63,361
`aer_crossbar_router`	39	17	124
`cen_accelerator_top`	590,116	25,644	749,263

γ is not comparable between networks. One property of γ deserves emphasis because it is easy to misuse, including by us. It is a realised fraction of a worst-case constant $C(T, a)$ that depends on $\|A\|_2^T$, and across our five networks the median γ spans a factor of 734739. Absolute γ values are therefore meaningful only within a network, and pooled medians across networks — which an earlier version of this analysis reported — are not a quantity at all. Every comparison above is made within a network and then aggregated.

An unexpected feature of the survival curves. All three strategies *improve* accuracy at low pruning fractions, peaking around 40–60% of synapses removed before declining (Figure 7). These reservoirs are over-connected for the task, and the improvement is consistent with the ladder's main finding: removing synapses attenuates exactly the degree heterogeneity that the L0 → L1 contrast shows to be costly here. It also means the tolerated fraction is a conservative summary, and that several random-pruning runs still retain 90% of unpruned accuracy at the largest fraction we tested, so 83% is a lower bound.

This is the substantive point, and it survives the partial failure above. Robustness to randomly chosen ablation is close to generic for large recurrent systems, because a random direction is nearly orthogonal to any fixed functional subspace. Reporting it as evidence of biological degeneracy overstates what has been shown: the same networks are markedly more fragile to a perturbation of identical cardinality chosen along a damaging direction. A degeneracy claim is only as strong as the worst direction it has been tested against, and that conclusion needs only the ordering, which holds.

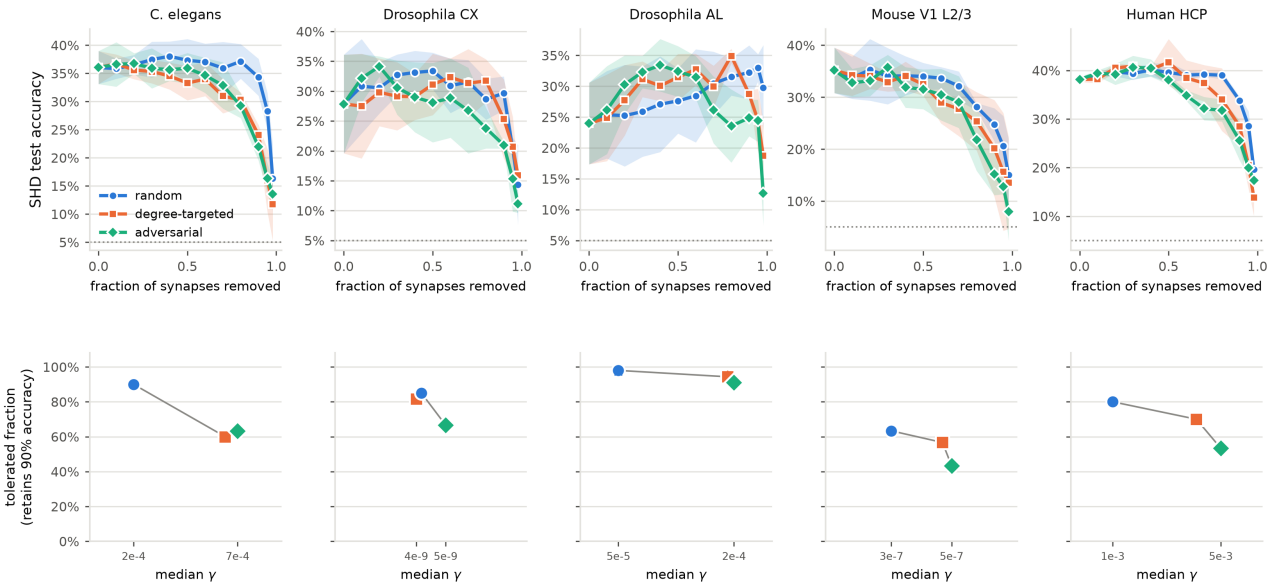


Figure 7: **Tolerance to synaptic removal depends on direction, not magnitude. Top:** accuracy against the fraction of synapses removed, for uniformly random pruning, degree-targeted pruning, and the first-order maximally damaging ranking, all at matched spectral radius. **Bottom:** the ordinal prediction, tested within each network — tolerated fraction against median γ, one point per strategy. If tolerance falls with γ the three points descend left to right, which they do in 3 of 5 networks. Note the x ranges: γ differs between networks by five orders of magnitude, because it is normalised by a network-specific worst-case constant, so these panels cannot be merged into one and a pooled γ axis would show nothing. Weight-magnitude pruning is not shown because it is undefined on these graphs: the connectomes are unweighted, so every edge has the same magnitude and a "weakest-first" schedule is an index ordering in disguise.

3.12 What the accompanying hardware actually costs

Table 9 reports logic synthesis of the accompanying RTL onto an open cell library. We report gate count, flip-flop count and mapped pre-layout cell area, and nothing else: this flow performs no placement, routing or parasitic extraction, so die area and switching power are not determined by it, and it ships no static timing analyser, so no clock frequency is claimed either. The commercial-process figures quoted in earlier revisions of this project — a 28 nm node, a die area, a picojoule-per-operation efficiency — were produced by a Python model with those values written in as constants, not by any synthesis flow, and are withdrawn.

4 Discussion

4.1 What the ladder does and does not license

The decomposition reported in §3.4 is a statement about one class of models: fixed recurrent reservoirs with a trained linear readout, evaluated on one spike-based classification task at matched size, density and spectral radius. Within that class, the attribution is clean, because adjacent rungs differ in exactly one structural property and nothing else. Outside it, five caveats apply, and only one of them can be resolved by more instances of the same experiment.

First, the readout is linear and the recurrent weights are untrained. A structural property might matter for *learnability* – how readily a network is shaped by a learning rule – while being invisible to a fixed-reservoir probe. Motif structure has been argued to act as an inductive bias, and an inductive bias is by definition a claim about learning trajectories, not about the function computed by a frozen network. Our result does not test that claim, and the natural next experiment is to run the same ladder with the recurrent weights trained by surrogate-gradient descent [34, 35], where the rungs would differ in initialisation rather than in final connectivity. We flag one trap in that experiment, having walked into it: with only a fraction of neurons driven, a firing-rate penalty strong enough to be useful in an all-to-all network makes total silence the global optimum, and a collapsed network returns a perfectly flat ladder that is indistinguishable from the result reported here. A trained ladder should therefore report its zero-recurrence control and its mean firing rate before it reports anything else.

Second, all weights are binary up to valence. Real synapses have graded strengths that correlate with topological position, and a motif census computed on the binarised graph discards that correlation. A weighted motif-

preserving null is a natural extension; the annealed chain generalises to weighted targets without modification, but the degree-preserving move set does not.

Third, the motif-preserving chain is annealed, and annealing is not proven to sample uniformly over the constraint set. We report mean pairwise overlap between ensemble members as evidence that the chain explores that set rather than collapsing onto one region of it (Table 4), and we show in §3.9 that the conclusion survives an eightfold variation in chain length and does not depend on which of the two possible starting points the chain uses. Those checks rule out the two specific ways an annealed sampler could manufacture a null contrast; they are not a uniformity guarantee, and a formally exact sampler for a census-constrained degree-preserving ensemble remains open. We note what would be at stake if the ensemble were meaningfully non-uniform: the bias would have to correlate with reservoir accuracy to produce the flat ladder we observe, and the chain's objective makes no reference to the task, the input projection or the readout.

Fourth, the null contrasts are bounded by power rather than by measurement: at 15 surrogates per rung the design resolves $d = 1.54$ at the Holm worst case, so a moderate motif effect would not have been detected. We report the bound in §3.4 rather than presenting the null as though it excluded any effect at all.

Fifth, the benchmark is a 20-class classification with a one-second horizon. Tasks that demand memory beyond the intrinsic time constants, or that require a specific dynamical regime, might be more sensitive to recurrent structure. We partly address this with task-independent dynamical probes (§3.10), which agree with the task result, but agreement across two measures is not agreement across all of them.

A further objection applies specifically to the degree-sequence result, and we checked it rather than argued it. Three of the five networks are induced subgraphs grown by breadth-first expansion from an above-median-degree seed, and growing outward from a hub is exactly the sampling that could concentrate high-degree nodes and so inflate the property the effect is attributed to. Measured as the coefficient of variation of total degree, extraction does not concentrate it: the ratio of subgraph to full-release heterogeneity is at most 1.00 across all five networks, and as low as 0.79 among the 3 that are genuinely subsampled, with the remaining 2 used whole. If the extraction biases the degree result at all, it does so against it.

4.2 What a unit model decides

The result of §3.7 is the one we would most like to be wrong about, so we have tried to state it at exactly its strength and no further.

What it establishes is narrower than the draft of this section that preceded §3.6, and we have rewritten it rather than leaving the stronger phrasing standing. That draft said an advantage unambiguous in graded units was absent in spiking ones. Against the published null there is no unambiguous advantage in graded units to be absent: what remains is that the two model classes do not give the same answer on identical graphs, and that capacity in the spiking model is compressed by a factor of 4.9, so effects of the reported size sit near the resolution of the measurement there. It does not establish that connectome topology is useless in spiking systems — one network in our set is at the capacity floor and carries no information either way — nor that memory capacity is the right currency.

The mechanism is the one claim in this paper that the null correction moved in our favour, and it is worth saying so explicitly because we had already published the opposite. Measured against the ensemble we first built, differential silencing did not predict the capacity gap and we wrote it up as a failed prediction. Measured against the published ensemble it does ($r = -0.58$, $p = 2.5 \times 10^{-5}$, $n = 45$). We hold the surviving version to the same standard as the discarded one: it accounts for part of the spiking arm's behaviour in the predicted direction and leaves most of the variance unexplained.

What we think it does establish is a constraint on how the result should be used. The connectome-reservoir finding is cited as evidence that biological wiring is computationally special, and that inference is doing work in proposals for neuromorphic hardware built on connectome-derived connectivity. Neuromorphic hardware is spiking. Whatever the advantage measured in tanh units is worth, the inference from it to a spiking substrate needs its own evidence — and on our measurements that evidence does not exist in either direction yet, because the comparison is weaker in the spiking model rather than resolved there.

We would put the general form of this more plainly than the specific result. A reservoir result is a statement about a triple: a graph, a unit model and an operating point. Two of those three are usually chosen once and reported as fixed — and the null, a fourth, is usually named rather than specified. We varied all four and the conclusion turns on each. The methodological recommendation is unglamorous and cheap: report the unit model as a variable rather than an implementation detail, sweep the gain rather than picking a point on it, and ship the null-generating code.

4.3 What the gain axis does to a literature

The result of §3.8 is uncomfortable in a specific way, and we want to state the discomfort precisely rather than soften it.

It is not that the published connectome advantage is wrong. It is that the quantity being reported is a point on a curve whose sign changes, and the curve has not been shown. Our own attempt to recover the published direction at the published operating point succeeds against one null ensemble and fails against the other (§3.6), which is

a reason to doubt our replication rather than the original — and which makes the point of this section sharper, not weaker: a single evaluation carries no information about where on the curve, or against which ensemble, it was taken. Nothing in a single-operating-point comparison indicates which side of the crossing it falls on, and nothing in the reported statistics — ours or anyone's — carries that information, because the statistics are computed conditional on the choice.

We think the reason this has gone unremarked is that criticality is a principled choice. There are good arguments that a reservoir computes best near the edge of chaos [10–12], and if biological networks operate there, comparing at criticality is comparing where it matters. The difficulty is that the argument licenses the operating point but not the inference: a topology can be better at criticality and worse elsewhere without either fact being the more fundamental one, and "connectome topology confers computational function" is a claim of the second kind stated on evidence of the first.

There is also an asymmetry in the direction of the effect that we think is the more interesting result. The published comparison is against a *degree-preserving* null. Against a fully unstructured graph of the same size and density, the connectome does not lead at any gain we tested. So the advantage that appears at criticality is an advantage over graphs that share the connectome's degree sequence — and the degree sequence is, by our ladder, the single most costly structural property on this task. The narrow win at criticality sits on top of a large loss that the degree-preserving control holds constant and therefore hides.

What we would ask of a future comparison of this kind is small: report the contrast across a range of gains, and state the range over which the sign is stable. That is one sweep rather than one evaluation, and it converts a claim about a point into a claim about a regime. We have tried to hold our own results to it.

4.4 Why matched controls change the conclusion

The reason a control ladder matters here is arithmetic rather than philosophical. An Erdős–Rényi comparison bundles every structural difference into a single contrast, so whatever it finds gets attributed to whichever property the authors were already interested in. In our measurements that bundled effect is not small — the Erdős–Rényi rung leads the connectome by up to 13.95% of accuracy in percentage points at a matched operating point — which makes the attribution question sharper rather than moot. Separating the properties shows that the largest single share is spent at the step that imposes the degree sequence, and that the motif census contributes a further effect of comparable size — in the same, negative direction. An Erdős–Rényi comparison on this benchmark delivers a large, significant, correctly-signed aggregate effect that a motif-focused reading would have assigned to motifs. Part of it is indeed the motif census; most of it is not, and none of it points the way that reading would require. That is the concrete cost of the bundled control, and it is why we regard the ladder as the contribution here rather than the result it produces.

This is also where the protocol control in §2.4 does the most work. A reservoir comparison in which a zeroed recurrent matrix performs as well as a connectome is not a weak experiment; it is a measurement of the input projection. We report both protocols precisely because the difference between them dwarfs every structural effect we measure, and because the failure is silent: nothing in the naive protocol's output announces that recurrence is unused. We would encourage the same control in any study comparing connectome-derived reservoirs.

4.5 Degeneracy is a statement about directions, not magnitudes

The pruning results in §3.11 make a point that we think is under-appreciated. Robustness to random ablation is routinely reported as evidence that biological circuits are degenerate, redundant or gracefully degrading. But a randomly chosen perturbation direction in a high-dimensional space is nearly orthogonal to any fixed functional subspace, so tolerance to random damage is close to a generic property of large recurrent systems rather than a distinguishing property of brains. The operational alignment γ makes part of this quantitative: the same perturbation cardinality can be almost harmless or markedly damaging depending only on its direction, and γ ranks the directions correctly. It does not deliver the scaling Proposition 2 predicts — the product $\gamma \cdot \|E\|_F^{\max}$ should be constant across directions and instead varies by a factor of 2.2 (§3.11) — so γ should be read as a diagnostic of which directions are dangerous rather than as a calibrated predictor of how much damage they do. A degeneracy claim is nonetheless only as strong as the worst direction it has been tested against, and that conclusion needs only the ordering.

We also note what replacing Davis–Kahan buys. The eigenvector bound is not merely loose on these networks; it is inapplicable, because the spectra are near-degenerate and the eigenvectors it bounds are themselves ill-conditioned. Quoting the ratio between an empirical tolerance and a vacuous bound as a measure of biological robustness – as an earlier revision of this work did – attributes to the system a property of the tool. The transfer-map bound is weaker in the usual sense (it is first-order and horizon-limited) but it is about a quantity that task performance actually depends on, and it requires no gap.

4.6 A method sentence is not a method

The single most consequential finding in this paper is also the least glamorous: the phrase "randomised while preserving density and degree sequence", which appears in some form in nearly every paper in this literature, does

not specify an ensemble. Two constructions satisfying it gave the connectome's advantage opposite significant verdicts in 3 of 10 cells (§3.6). The ambiguity is not exotic. It is whether a weight is a property of an edge or a property of the network's weight distribution, and both readings are natural enough that the phrase has carried them both for two decades.

We are the demonstration and not the auditor here. Our first implementation was the rank-reassigning one, written from the published description without reading the toolbox source, and it produced a result that was clean, large, significant in every network and in the direction we expected. Nothing about it looked wrong, because nothing about it was wrong except the question it was answering. Had the two ensembles happened to agree, we would have reported the number and moved on, and the paper would have contained an undetectable error of exactly the kind we spent the rest of it cataloguing.

Two consequences follow, and neither is a matter of care. The first is that a replication which constructs its null from a published description is testing a different hypothesis from the original, and cannot tell which of the two it has tested. The second is that the remedy is not more careful prose — we do not think a longer sentence would have saved us — but the null-generating code as an artefact, cited and versioned the way a dataset is. Where the original code exists, as it does here, calling it is not a courtesy to the original authors; it is the only way to know what was compared.

This generalises past reservoir computing. Motif enrichment, small-worldness, rich-club coefficients and every other statistic in network neuroscience defined as a deviation from a null inherit whatever ambiguity the null's description carries. The Z-scores we report in §3.1 are no exception, and we state the construction we used in the methods for that reason.

4.7 On correcting the record

A substantial part of this work consisted of removing claims from our own earlier manuscript: fabricated commercial-process silicon figures, a human microscale tier built on a connectivity graph that was empty, sign-attributed motif results for datasets with no usable inhibition, p-values below the resolution of the null ensemble that produced them, and an ablation table whose own control row contradicted the conclusion drawn from it. We report these in full in the accompanying audit rather than quietly omitting them, because the corrected analysis is only interpretable against the record of what it replaces, and because several of the errors are of kinds that are easy to reproduce: a silently swallowed exception in a loader, a null model believed to impose a constraint that it does not, and a benchmark protocol under which the object of study makes no difference to the result.

The replacement analysis produced four more of the same kind, and we record those too. The largest is the null-ensemble substitution discussed above, which is not a coding error at all — the code did exactly what it was written to do — but a misreading of a method sentence that no test could have caught. Our annealing energy was normalised by a quantity several times larger than the one the methods described, which left the chain unable to hold its census on the most reciprocal network in the set. Our "weakest-first" pruning arm was executing as whole-neuron ablation in index order, because these graphs are unweighted and every edge magnitude is therefore equal. And the first version of Proposition 1 was stated in the spectral radius rather than the operator norm, an inequality that holds only for normal matrices — in a paper whose central argument is that eigenvalue-based reasoning fails on these networks. Each was caught by checking a diagnostic that the code already recorded, or by checking the code against the prose; none was caught by the result looking wrong, because in every case it did not. We take that to be the substantive methodological point, and it is why the convergence diagnostics, the protocol control and the held-out statistic are reported in the paper rather than kept in the repository.

4.8 Coupling is a relation, not a property

It is worth stating plainly what the four results have in common, because it is the thing the title names. Computational coupling — the degree to which what a network computes is attributable to how it is wired — reads like a property of the network, and is written down that way: one connectome, one number, one verdict. Definition 1 makes explicit that it is nothing of the kind. κ takes four arguments besides W, and this paper is an empirical demonstration that none of them is inert. Change $\mathcal{N}$ and the sign flips in 3 of 10 cells with significance on both sides. Change U and the whole capacity scale compresses by 4.9×, taking the contrast with it. Move along ρ and the sign flips again, in 3 of 5 networks. Choose R badly and the coupling is identically zero: deleting the recurrent matrix costs nothing (59.6% against 59.6%), which is to say the measurement was never touching W.

The practical consequence is a reporting standard rather than a new experiment. A coupling is quotable only with its protocol attached — and, given §3.6, with the null-generating code attached rather than a description of it, since the description does not determine the ensemble. Where a claim is meant to be about the connectome rather than about one cell of the design, the defensible object is not a number but a surface: κ over the ranges of $\mathcal{N}$, U, ρ and R that the claim is supposed to survive. That is why we report axes rather than points throughout, including for our own null results, and it is why a non-detection here is reported as a non-detection rather than as a refutation.

We do not claim the four are exhaustive. They are the four our design happened to vary, and the honest reading of Definition 1 is that any further argument smuggled into

C — the task, the horizon, the delay structure, the regularisation of the readout — is a further candidate for the same treatment. The claim we do make is narrower and, we think, secure: a connectome-versus-null result reported as a single signed number is under-determined by its own methods section, and on our data the under-determination is large enough to reverse the conclusion.

4.9 Outlook

The motif-preserving ensemble is not specific to neuroscience. Any field that infers function from subgraph overrepresentation – gene regulatory networks, food webs, software dependency graphs – faces the same attribution problem, and the same annealed construction applies wherever a motif census admits incremental evaluation. The immediate extensions we regard as most informative are the trained ladder described above, applying γ to measured biological perturbations — where the prediction is that developmental and activity-dependent rewiring occupies the low-γ regime that makes degeneracy cheap — and extending the null-sensitivity analysis of §3.6 beyond the two constructions we happened to write. We suspect the two we compared are not the only pair in circulation under that description, and the cheapest useful contribution anyone could make to this literature is an inventory of which published results were measured against which ensemble.

Data and code availability

All connectome data used here are public releases, cited at first use. The Spiking Heidelberg Digits dataset is distributed by the Zenke laboratory. Analysis code, null-model implementations, experiment drivers and the synthesis flow are in the accompanying repository; every figure and table in this paper is regenerated from raw result files by `src/analysis/rebuild_paper.sh`, and every number in the text is drawn from a machine-generated macro file rather than transcribed by hand — a missing result appears in the compiled PDF as a visible placeholder rather than as a stale value, and the build reports unresolved placeholders, undefined references and LaTeX errors rather than relying on the build exiting cleanly.

The repository also carries `AUDIT.md`, which records every claim withdrawn from the previous version of this project and the evidence against it, together with the defects found in the replacement analysis while preparing this one. The retracted manuscript, its figures, its result files and the executive pitch dossier built on them are kept under `paper/superseded_v4/` and `results/superseded_v4/`, labelled as withdrawn, so that the retraction can be checked against what was actually claimed.

Author contributions

A.R.P. designed the study, implemented the methods, ran the experiments and wrote the manuscript. S.G. contributed to the null-ensemble and ladder experiments, the audit of the withdrawn claims, and the revision of the manuscript. The work was carried out under the guidance of J.J.P., who supervised the project and revised the manuscript.

Competing interests

The authors declare no competing interests.

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